

Weil Classes and the Hodge Conjecture
Abelian Varieties, Algebraic Cycles and the Variational Problem

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Draft, October 2026

Preface

This book is about one question and one family of examples. The question is the Hodge conjecture: on a smooth complex projective variety, is every rational cohomology class of type (p, p) a rational combination of classes of subvarieties? The examples are abelian varieties, and among them the abelian varieties of Weil type. That is where the first Hodge classes appear that divisors cannot produce.

The conjecture is not proved here. Nobody has proved it, and a reader who opens the book hoping for a proof should know that at the start. What the book does is narrower and, I hope, more useful. It explains why the conjecture is true where it is known to be true, with complete proofs where the arguments are short enough to give. It explains what changes when one moves from divisors to classes of higher codimension. And it identifies, as precisely as current knowledge allows, the statement that is still missing for abelian varieties. For a general abelian variety of Weil type the whole conjecture comes down to the algebraicity of a single explicit class. That reduction is classical in spirit, since it goes back to Weil, but its consequences are worth following carefully, and most of the book does so.

The first three chapters are background. Chapter 1 states the conjecture, explains why the hypotheses are what they are, and lists the counterexamples to its stronger variants. Chapter 2 collects the Hodge theory the rest of the book uses. Chapter 3 proves the theorem of Lefschetz on $(1, 1)$ classes, which settles degree two. The proof is short, and the way it fails in higher degree explains a good deal of the difficulty.

Chapters 4 and 5 are the core of the book. The cohomology of an abelian variety is an exterior algebra, its Hodge classes are the invariants of a reductive group, and the question becomes one of representation theory followed by one of geometry. Weil's classes are the first case where the representation theory says yes and the geometry has not caught up. Chapter 6 records the cases in which it has: the work of Schoen, Koike and Markman, the members where divisors suffice, and the more recent preprints, kept clearly separate from refereed theorems. Chapter 7 is an attempt at the remaining case. It follows the natural route, deforming a cycle from a special member of the family to a general one, and shows exactly where that route stops.

The reader I have in mind knows the basic language of algebraic geometry and complex manifolds at the level of Griffiths and Harris or Voisin's two volumes. Where a proof would need more than a few pages of machinery that is well covered elsewhere, I state the result precisely, give a reference, and say what the proof uses.

Some of the material on Weil classes and their algebraic loci first appeared in my preprint *Algebraic Loci of Weil Classes on Abelian Varieties* [BhH8]. Most of what is taken from it is proved again here. Where a proof is only sketched, the text says so and points to the preprint. The arguments are analytical throughout: no step rests on a computer calculation. Every

place where a result depends on an unrefereed preprint by someone else is marked as such.

A note on preparation

Drafts of parts of this book were prepared with the help of an AI language model, used to draft and revise text under the author's direction. Every statement and proof has been checked by the author, who is responsible for the content.

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Chapter 1

The conjecture

1.1 Hodge classes

Let X be a smooth complex projective variety of dimension d . Its cohomology with rational coefficients carries two structures that have, on the face of it, nothing to do with each other. The first is the rational structure itself, the lattice $H^k(X, \mathbb{Z})$ modulo torsion inside $H^k(X, \mathbb{C})$, which is a topological invariant. The second is the Hodge decomposition

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X), \quad \overline{H^{p,q}(X)} = H^{q,p}(X), \quad (1.1)$$

in which $H^{p,q}(X)$ is the space of classes represented by closed forms of type (p, q) . The decomposition depends on the complex structure and moves when the complex structure moves. The rational structure does not.

The Hodge classes are the classes where the two structures meet.

Definition 1.1. For $0 \leq p \leq d$ the space of *rational Hodge classes of degree $2p$* is

$$\text{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X),$$

the intersection being taken inside $H^{2p}(X, \mathbb{C})$.

Because the decomposition is preserved by cup product, in the sense that $H^{p,q} \cup H^{p',q'} \subseteq H^{p+p',q+q'}$, the direct sum $\text{Hdg}^*(X) = \bigoplus_p \text{Hdg}^p(X)$ is a graded $\mathbb{Q}$ -subalgebra of $H^{\text{even}}(X, \mathbb{Q})$. It is called the *Hodge ring* of X .

There is no reason, a priori, for $\text{Hdg}^p(X)$ to be anything but zero. For a very general complex structure on a given manifold the space $H^{p,p}$ is in general position with respect to the rational lattice, and the intersection is as small as it can be. What forces it to be larger is geometry.

1.2 Classes of subvarieties

Let $Z \subseteq X$ be a closed irreducible subvariety of codimension p . Integration over the smooth part of Z defines a linear form on closed $(2d - 2p)$ -forms,

$$\alpha \longmapsto \int_{Z_{\text{reg}}} \alpha,$$

which converges and vanishes on exact forms. This is a theorem of Lelong for the convergence and of Stokes, applied on a resolution of Z , for the vanishing. By Poincaré duality the form corresponds to a class $[Z] \in H^{2p}(X, \mathbb{Q})$. In fact the class lies in the image of $H^{2p}(X, \mathbb{Z})$, since it is the Poincaré dual of the fundamental class of Z in Borel–Moore homology. Extending linearly to cycles, that is, to formal combinations $\sum n_i Z_i$, and passing to rational coefficients gives the *cycle class map*

$$\mathrm{cl}^p: Z^p(X)_{\mathbb{Q}} \longrightarrow H^{2p}(X, \mathbb{Q}).$$

It factors through rational equivalence, so it is defined on the Chow group $\mathrm{CH}^p(X)_{\mathbb{Q}}$.

Proposition 1.2. *For every cycle Z of codimension p the class $\mathrm{cl}^p(Z)$ lies in $\mathrm{Hdg}^p(X)$.*

Proof. The class is rational by construction, so it suffices to show that it has type (p, p) . Under Poincaré duality, $H^{2p}(X, \mathbb{C})$ is dual to $H^{2d-2p}(X, \mathbb{C})$, and the summand $H^{p,p}$ is the annihilator of every summand $H^{r,s}$ with $r+s = 2d-2p$ except $H^{d-p,d-p}$. So we must show that $\int_Z \alpha = 0$ for every closed form α of type (r, s) with $r+s = 2d-2p$ and $r \neq d-p$. But Z_{reg} is a complex manifold of dimension $d-p$, and the restriction of a form of type (r, s) to it is of type (r, s) there. A form of type (r, s) on a complex manifold of dimension $d-p$ vanishes as soon as $r > d-p$ or $s > d-p$. Since $r+s = 2(d-p)$ and $r \neq d-p$, one of the two inequalities holds. $\square$

So subvarieties give Hodge classes. The conjecture says that they give all of them.

Conjecture 1.3 (Hodge). *Let X be a smooth complex projective variety. For every p the cycle class map*

$$\mathrm{cl}^p: \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X)$$

is surjective.

Hodge raised the question in his address to the International Congress of 1950, in a form with integral coefficients. The formulation above, with rational coefficients, is the one that survived. The next sections explain why each of its hypotheses is needed.

1.3 Why rational coefficients

One might ask whether every integral class of type (p, p) , that is, every element of $H^{2p}(X, \mathbb{Z})$ whose image in $H^{2p}(X, \mathbb{C})$ lies in $H^{p,p}(X)$, is the class of an integral cycle. In degree two the answer is yes; this is the content of Chapter 3. In higher degree it is no.

Atiyah and Hirzebruch [AH62] found torsion classes that are not algebraic. Their argument is topological. The class of an algebraic cycle is the pushforward of a fundamental class along a resolution, and such classes are annihilated by certain odd-degree Steenrod operations. They then exhibited smooth projective varieties with torsion classes, which are automatically of type (p, p) , on which those operations do not vanish. Totaro [Tot97] later explained the obstruction through complex cobordism, and found further examples.

Torsion classes are invisible rationally, so one might hope that the integral statement holds modulo torsion. It does not. Kollár [Kol92] showed that on a very general hypersurface $X \subset \mathbb{P}^4$ of sufficiently large degree m , with m divisible by a high power of a prime, every curve has degree divisible by that power. Yet the generator of $H^4(X, \mathbb{Z}) \cong \mathbb{Z}$, which is of type $(2, 2)$ for dimension reasons, has degree 1. So the generator is not the class of a curve, although m times

it is, namely the class of the intersection of X with a plane. Rationally nothing is lost, and integrally the conjecture fails.

1.4 Why projective

The definitions make sense for every compact Kähler manifold, and the Hodge decomposition holds there. One could ask whether every rational (p, p) class on a compact Kähler manifold is a combination of classes of analytic subvarieties, or more generously of Chern classes of coherent sheaves. For degree two, the theorem of Lefschetz holds in the Kähler setting for classes that are Chern classes of line bundles. But a general complex torus has no line bundles beyond the topologically trivial ones, and no analytic subvarieties of positive dimension at all, while it may still have Hodge classes in higher degree.

So the analytic version of the conjecture fails as soon as such a torus has a nonzero Hodge class of degree four, and such tori exist. Voisin [Voi02b] went much further. She constructed complex tori, of Weil type in the sense of Chapter 5, carrying nonzero Hodge classes of degree four, on which every coherent sheaf has vanishing Chern classes in that degree. So the Kähler analogue fails even with Chern classes of sheaves in place of subvarieties, and projectivity is used essentially. We return to Voisin's tori in Section 5.5, because they are the same Weil classes this book is about, seen from the other side.

1.5 What is known in general

The conjecture is trivial in degrees 0 and $2d$. In degree two it is the theorem of Lefschetz (Chapter 3). The hard Lefschetz theorem then gives it in degree $2d - 2$ (Corollary 2.10). In particular it holds for every variety of dimension at most three. For fourfolds the first open degree is four, and there it is open in general, though known for many classes of fourfolds, among them uniruled ones [CM78]. Beyond that there is no general method. The known results are proved family by family, and abelian varieties are the family about which the most is known.

A useful reformulation is due to Mumford and Deligne. The rational cohomology $H^k(X, \mathbb{Q})$ with its Hodge decomposition is a rational Hodge structure of weight k , and every Hodge structure has a symmetry group, the *Mumford–Tate group*. It is the smallest algebraic subgroup of $\mathrm{GL}(H^k(X, \mathbb{Q}))$ defined over $\mathbb{Q}$ whose real points contain the image of the homomorphism that defines the Hodge decomposition (Section 2.3). Its variant the *Hodge group* has the property that the Hodge classes in tensor constructions on $H^k(X, \mathbb{Q})$ are exactly its invariants. For an abelian variety A the whole cohomology is the exterior algebra on $H^1(A, \mathbb{Q})$, and so the Hodge ring of A is the ring of invariants of a single reductive group acting on an exterior algebra. That is why abelian varieties are tractable: the Hodge classes can be computed. The difficulty, once they are computed, is to find the cycles.

1.6 The plan

The first half of the book makes the last two sentences precise. Chapter 2 recalls the Hodge-theoretic tools, Chapter 3 proves the theorem in degree two, and Chapter 4 computes the

cohomology of abelian varieties and explains how the Hodge ring is governed by the Hodge group.

Chapter 5 introduces Weil's examples. Let K be an imaginary quadratic field acting on an abelian variety A of dimension $2n$, in such a way that the two eigenspaces of K on the tangent space both have dimension n . Then $H^{2n}(A, \mathbb{Q})$ contains a two-dimensional space of Hodge classes, the Weil classes, which for a general such A are not products of divisor classes. We prove that for a general member of a family of such varieties, the Hodge ring is spanned by powers of the polarisation together with this plane. So the Hodge conjecture for the general member is the algebraicity of one class.

Chapter 6 surveys the cases in which this class is known to be algebraic, and Chapter 7 attempts the general case.

Chapter 2

Hodge theory: the tools we use

This chapter collects what the rest of the book needs. Most of it is standard and is proved in Voisin's books [Voi02, Voi03] or in Griffiths and Harris [GH78]. We give proofs when they are short and when the argument itself matters later.

2.1 Hodge structures

Definition 2.1. A *rational Hodge structure of weight k* is a finite-dimensional $\mathbb{Q}$ -vector space V together with a decomposition

$$V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C} = \bigoplus_{p+q=k} V^{p,q}, \quad \overline{V^{p,q}} = V^{q,p}.$$

A *morphism* of Hodge structures of the same weight is a $\mathbb{Q}$ -linear map whose complexification preserves the decomposition.

The Hodge filtration $F^p V_{\mathbb{C}} = \bigoplus_{p' \geq p} V^{p',k-p'}$ determines the decomposition, through $V^{p,q} = F^p \cap \overline{F^q}$. In families it is the filtration, not the decomposition, that varies holomorphically.

Direct sums, tensor products, duals and exterior powers of Hodge structures are again Hodge structures, with the obvious types; for example $(V \otimes W)^{p,q} = \bigoplus V^{a,b} \otimes W^{p-a,q-b}$, and the dual of a structure of weight k has weight $-k$, with $(V^{\vee})^{p,q}$ dual to $V^{-p,-q}$. The *Tate structure* $\mathbb{Q}(1)$ is the one-dimensional structure of weight -2 and type $(-1, -1)$, and $V(m) = V \otimes \mathbb{Q}(1)^{\otimes m}$ shifts the weight by $-2m$ without changing anything else.

Definition 2.2. A *Hodge class* in a Hodge structure V of even weight $2p$ is an element of $V \cap V^{p,p}$. Equivalently, it is a morphism of Hodge structures $\mathbb{Q}(-p) \rightarrow V$.

For a smooth projective X , the cohomology $H^k(X, \mathbb{Q})$ is a Hodge structure of weight k by the Hodge decomposition, and $\text{Hdg}^p(X)$ is the space of Hodge classes in $H^{2p}(X, \mathbb{Q})$. The second description is the useful one: a Hodge class is the same thing as a morphism of Hodge structures, and the Hodge conjecture predicts that such morphisms come from geometry.

2.2 Polarisation

Let V be a Hodge structure of weight k . The operator C that acts on $V^{p,q}$ by i^{p-q} is defined over $\mathbb{R}$, and it is called the Weil operator.

Definition 2.3. A *polarisation* of V is a bilinear form $Q: V \times V \rightarrow \mathbb{Q}$, symmetric if k is even and alternating if k is odd, such that the complexification satisfies $Q(V^{p,q}, V^{p',q'}) = 0$ unless $p = q'$, and such that the hermitian form $Q(Cx, \bar{y})$ is positive definite on $V_{\mathbb{C}}$.

Proposition 2.4. Let V be a polarised Hodge structure and $U \subseteq V$ a sub-Hodge structure, that is, a subspace with $U_{\mathbb{C}} = \bigoplus (U_{\mathbb{C}} \cap V^{p,q})$. Then the orthogonal complement $U^{\perp}$ with respect to Q is a sub-Hodge structure, and $V = U \oplus U^{\perp}$. In particular, polarisable Hodge structures form a semisimple category.

Proof. The form $h(x, y) = Q(Cx, \bar{y})$ is positive definite, so $U_{\mathbb{C}}$ and its h -orthogonal complement span $V_{\mathbb{C}}$ and meet in zero. Since C preserves $U_{\mathbb{C}}$, the h -orthogonal complement of $U_{\mathbb{C}}$ equals the Q -orthogonal complement of $\overline{U_{\mathbb{C}}} = U_{\mathbb{C}}$, which is $(U^{\perp})_{\mathbb{C}}$. So $V = U \oplus U^{\perp}$ over $\mathbb{Q}$. The Q -orthogonal of a subspace stable under the decomposition is stable under it, because Q pairs $V^{p,q}$ only with $V^{q,p}$. $\square$

The cohomology of a smooth projective variety is polarisable. This is the content of the Hodge–Riemann relations, recalled in Section 2.4. Semisimplicity is used constantly: it lets us split off the part of a cohomology group that we understand and look at what is left.

2.3 Mumford–Tate and Hodge groups

A Hodge structure of weight k on V is the same thing as a homomorphism of real algebraic groups

$$h: \mathbb{S} = \text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m \longrightarrow \text{GL}(V_{\mathbb{R}}),$$

where $z \in \mathbb{S}(\mathbb{R}) = \mathbb{C}^{\times}$ acts on $V^{p,q}$ by $z^{-p}\bar{z}^{-q}$. The weight is recorded by the restriction of h to $\mathbb{R}^{\times}$, and the Weil operator is $h(i)$.

Definition 2.5. The *Mumford–Tate group* $\text{MT}(V)$ is the smallest algebraic subgroup of $\text{GL}(V)$, defined over $\mathbb{Q}$, whose real points contain $h(\mathbb{S}(\mathbb{R}))$. The *Hodge group* $\text{Hg}(V)$ is the smallest one whose real points contain $h(U(1))$, where $U(1) \subset \mathbb{S}(\mathbb{R})$ is the unit circle.

For a structure of weight $k \neq 0$ the Mumford–Tate group is the product of the Hodge group and the homotheties. The Hodge group is the right object for counting Hodge classes.

Theorem 2.6. Let V be a rational Hodge structure and let $T = V^{\otimes a} \otimes (V^{\vee})^{\otimes b}$ be a tensor space, with its Hodge structure of some weight w . If w is even, then an element of T is a Hodge class if and only if it is invariant under $\text{Hg}(V)$. If V is polarisable, $\text{Hg}(V)$ is a connected reductive group.

Proof. On $T^{p,q}$ the circle acts through $z \mapsto z^{-p}\bar{z}^{-q} = z^{q-p}$, since $\bar{z} = z^{-1}$ on $U(1)$. So the fixed vectors of $h(U(1))$ in $T_{\mathbb{C}}$ are exactly the elements of $\bigoplus_p T^{p,p}$, which is $T^{w/2, w/2}$. An element $t \in T$ (rational) is fixed by $h(U(1))$ if and only if it is fixed by the smallest $\mathbb{Q}$ -algebraic subgroup containing $h(U(1))$, because the stabiliser of t is an algebraic subgroup defined over $\mathbb{Q}$. This is the first claim. The group is connected because $U(1)$ is. Reductivity uses the polarisation: the form $Q(Cx, \bar{y})$ makes the real points of a certain inner form of $\text{Hg}(V)$ compact, which forces reductivity. We refer to [Del82] for this step. $\square$

So the Hodge ring of X , in so far as it sits inside tensor constructions on one Hodge structure, is a ring of invariants. For abelian varieties all of the cohomology sits inside the exterior algebra on H^1 , and this observation is the starting point of Chapter 4.

2.4 The Kähler package

Let X be smooth projective of dimension d and let $\eta \in H^2(X, \mathbb{Q})$ be the class of an ample divisor. Cup product with η is the *Lefschetz operator* L . It is a morphism of Hodge structures $H^k(X, \mathbb{Q}) \rightarrow H^{k+2}(X, \mathbb{Q})(1)$, because η is of type $(1, 1)$.

Theorem 2.7 (Hard Lefschetz). *For $0 \leq k \leq d$ the map $L^{d-k}: H^k(X, \mathbb{Q}) \rightarrow H^{2d-k}(X, \mathbb{Q})$ is an isomorphism.*

Define the primitive part $P^k = \ker(L^{d-k+1}: H^k \rightarrow H^{2d-k+2})$. Hard Lefschetz gives the *Lefschetz decomposition*

$$H^k(X, \mathbb{Q}) = \bigoplus_{j \geq 0} L^j P^{k-2j},$$

and every summand is a sub-Hodge structure, up to Tate twist.

Theorem 2.8 (Hodge–Riemann bilinear relations). *For $k \leq d$ the form*

$$Q(x, y) = (-1)^{k(k-1)/2} \int_X \eta^{d-k} \cup x \cup y$$

is a polarisation of the Hodge structure P^k . Twisting by the appropriate signs on the summands of the Lefschetz decomposition gives a polarisation of $H^k(X, \mathbb{Q})$.

Both theorems are proved with harmonic forms; see [Voi02, Chapter 6].

2.5 Correspondences

By the Künneth formula and Poincaré duality,

$$H^{2d_X+2r}(X \times Y, \mathbb{Q}) \cong \bigoplus_k \operatorname{Hom}(H^k(X, \mathbb{Q}), H^{k+2r}(Y, \mathbb{Q}))$$

for smooth projective X, Y with $\dim X = d_X$. A class $\Gamma \in H^*(X \times Y, \mathbb{Q})$ acts as $\Gamma_*(x) = \operatorname{pr}_{Y*}(\Gamma \cup \operatorname{pr}_X^* x)$. Under this isomorphism the Hodge classes on $X \times Y$ correspond to morphisms of Hodge structures, up to Tate twist.

Proposition 2.9. *If Γ is the class of an algebraic cycle on $X \times Y$, then Γ_* carries classes of algebraic cycles on X to classes of algebraic cycles on Y .*

Proof. Pullback along the projection, intersection with a cycle and proper pushforward are all defined on Chow groups, and they are compatible with the cycle class map [Ful98, Chapter 19]. $\square$

This is the basic mechanism for moving algebraicity around: once a single correspondence is known to be algebraic, it transports algebraicity from one variety to another. Most known cases of the Hodge conjecture for abelian varieties are proved this way, starting from divisors.

2.6 Degree $2d - 2$

Corollary 2.10. *If the Hodge conjecture holds for X in degree 2, it holds in degree $2d - 2$. In particular, by the theorem of Lefschetz in Chapter 3, it holds in degree $2d - 2$ for every smooth projective X .*

Proof. The map $L^{d-2}: H^2(X, \mathbb{Q}) \rightarrow H^{2d-2}(X, \mathbb{Q})(d-2)$ is an isomorphism of Hodge structures by Theorem 2.7. An isomorphism of Hodge structures is a bijection on Hodge classes. So every Hodge class of degree $2d - 2$ has the form $\eta^{d-2} \cup \alpha$ with $\alpha \in \text{Hdg}^1(X)$. If α is the class of a divisor D , then $\eta^{d-2} \cup \alpha$ is the class of the curve $D \cdot H_1 \cdots H_{d-2}$ for general hyperplane sections H_j in the class of η . $\square$

The argument is worth looking at closely because it shows the general pattern. The Lefschetz operator L is algebraic, being intersection with a hyperplane. Its inverse on the image is not known to be algebraic. That is the Lefschetz standard conjecture, and if it were known, the Hodge conjecture in degree $2p$ for $p > d/2$ would follow from degree $2d - 2p$. The problem is to find inverses, not to find the forward map. This pattern will appear again in Chapter 7.

2.7 Families

Let $\pi: \mathcal{X} \rightarrow S$ be a smooth projective morphism with connected base. The groups $H^k(X_s, \mathbb{Q})$ form a local system $R^k \pi_* \mathbb{Q}$, and the Hodge filtrations vary holomorphically and satisfy Griffiths transversality. The result is a variation of Hodge structure. A Hodge class at one point need not remain a Hodge class when it is transported to nearby points. The set of nearby points where it does is called the *Hodge locus* of the class.

Theorem 2.11 (Cattani–Deligne–Kaplan [CDK95]). *For a variation of Hodge structure coming from a smooth projective family over a quasi-projective base, the Hodge locus of every class is a countable union of closed algebraic subvarieties of S . The locus of classes of bounded norm (with respect to the polarisation) is a finite union.*

The Hodge conjecture predicts this: if every Hodge class is algebraic, then the Hodge locus is the image of a countable union of relative Hilbert schemes, which are algebraic. Cattani, Deligne and Kaplan proved it with no conjecture at all. Their theorem is one of the best pieces of evidence for the Hodge conjecture.

Theorem 2.12 (Deligne, global invariant cycles [Del71]). *Let $\bar{\mathcal{X}}$ be a smooth projective compactification of $\mathcal{X}$. Then the classes in $H^k(X_s, \mathbb{Q})$ invariant under the monodromy of π are exactly the restrictions of classes in $H^k(\bar{\mathcal{X}}, \mathbb{Q})$. Moreover, the invariant classes form a sub-Hodge structure.*

This theorem gives the standard reduction of the variational question to the Hodge conjecture for the total space. Suppose a monodromy-invariant class is a Hodge class at one point. By the second part of the theorem it is then a Hodge class at every point. By the first part it is the restriction of a class on $\bar{\mathcal{X}}$, and since restriction is a surjective morphism of polarisable Hodge structures onto the invariant part, Proposition 2.4 lets us choose that class to be a Hodge class. If the Hodge conjecture holds for $\bar{\mathcal{X}}$, the class is algebraic at every point. The difficulty is that $\bar{\mathcal{X}}$ is a larger variety, about which we know less.

Chapter 3

Degree two: the theorem of Lefschetz

In degree two the Hodge conjecture is a theorem, and it holds even with integral coefficients. The proof takes a page. It is worth giving in full, because each of its three steps fails in higher degree for a different reason, and those failures explain why the general case is hard.

3.1 Statement

Theorem 3.1 (Lefschetz). *Let X be a smooth complex projective variety. Every class $c \in H^2(X, \mathbb{Z})$ whose image in $H^2(X, \mathbb{C})$ lies in $H^{1,1}(X)$ is the first Chern class of a line bundle, and therefore the class of a divisor. In particular $\text{Hdg}^1(X)$ is spanned by classes of divisors.*

3.2 Proof

Step 1: line bundles and the exponential sequence

On the complex manifold X there is an exact sequence of sheaves

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2\pi i} \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^\times \longrightarrow 0.$$

Isomorphism classes of holomorphic line bundles are the elements of $H^1(X, \mathcal{O}_X^\times)$, and the connecting map $\delta: H^1(X, \mathcal{O}_X^\times) \rightarrow H^2(X, \mathbb{Z})$ of the long exact sequence is the first Chern class, up to a sign fixed by the conventions [GH78, p. 139]. The next terms of the sequence are

$$H^1(X, \mathcal{O}_X^\times) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \xrightarrow{j} H^2(X, \mathcal{O}_X),$$

so an integral class is the Chern class of a holomorphic line bundle exactly when $j(c) = 0$.

Step 2: identifying j with the $(0, 2)$ part

The map j is induced by the inclusion of sheaves $\mathbb{Z} \rightarrow \mathcal{O}_X$, which factors as $\mathbb{Z} \rightarrow \mathbb{C} \rightarrow \mathcal{O}_X$. So j is the composite

$$H^2(X, \mathbb{Z}) \longrightarrow H^2(X, \mathbb{C}) \longrightarrow H^2(X, \mathcal{O}_X).$$

Under the Dolbeault isomorphism $H^2(X, \mathcal{O}_X) \cong H^{0,2}(X)$, the second map is the projection of the Hodge decomposition onto its $(0, 2)$ summand. To see this, represent a class in $H^2(X, \mathbb{C})$ by a closed form $\alpha = \alpha^{2,0} + \alpha^{1,1} + \alpha^{0,2}$ that is harmonic, so that each component is closed. The map to $H^2(X, \mathcal{O}_X)$, computed with the Dolbeault resolution of $\mathcal{O}_X$, keeps only the $(0, 2)$ component.

Now let c be an integral class of type $(1, 1)$. Its $(0, 2)$ component is zero, so $j(c) = 0$, and Step 1 gives a holomorphic line bundle L with $c_1(L) = c$.

The same argument shows something slightly stronger. For a class c with real image, the $(2, 0)$ and $(0, 2)$ components are complex conjugate. So $j(c) = 0$ already forces c to be of type $(1, 1)$, and the Chern classes of line bundles are exactly the integral classes of type $(1, 1)$.

Step 3: from line bundles to divisors

Here projectivity enters. By GAGA [Ser56], the holomorphic line bundle L on the projective variety X is algebraic. Let $\mathcal{O}_X(1)$ be very ample. By Serre's theorem, $L(m) = L \otimes \mathcal{O}_X(m)$ is generated by global sections for m large, and in particular it has a nonzero section, whose zero scheme is an effective divisor D_1 with $\mathcal{O}_X(D_1) \cong L(m)$. A general hyperplane section D_2 has $\mathcal{O}_X(D_2) \cong \mathcal{O}_X(m)$. So $L \cong \mathcal{O}_X(D_1 - D_2)$ and

$$c = c_1(L) = [D_1] - [D_2],$$

using the standard identity $c_1(\mathcal{O}_X(D)) = [D]$ [GH78, p. 141]. This proves the theorem. $\square$

3.3 Why the proof does not go further

Each step fails in higher codimension, in a different way.

There is no exponential sequence for cycles of codimension $p \geq 2$

The first step used that line bundles, which are cohomology classes of $\mathcal{O}_X^\times$, are classified by a group that sits in a long exact sequence with $H^2(X, \mathbb{Z})$. The analogue in codimension p is Deligne cohomology $H_{\mathcal{D}}^{2p}(X, \mathbb{Z}(p))$. It sits in an exact sequence

$$0 \longrightarrow J^p(X) \longrightarrow H_{\mathcal{D}}^{2p}(X, \mathbb{Z}(p)) \longrightarrow \text{Hdg}^p(X)_{\mathbb{Z}} \longrightarrow 0,$$

where $J^p(X)$ is Griffiths' intermediate Jacobian, and every cycle of codimension p has a class in it. For $p = 1$, Deligne cohomology is $H^1(X, \mathcal{O}_X^\times)$, the Picard group, and this sequence is the one of Step 1. For $p \geq 2$ the analogous sequence exists and is exact, but it says nothing about cycles: there is no reason for an element of Deligne cohomology to come from a cycle. Indeed the image of the cycle map in $J^p(X)$, the image of the Abel–Jacobi map, can be much smaller than $J^p(X)$, as Griffiths showed. Whatever proves the Hodge conjecture cannot be a cohomological lifting argument of this kind.

Step 2 has no analogue that picks out cycles

In degree two, being of type $(1, 1)$ is equivalent to vanishing of the $(0, 2)$ part, a single condition that the exponential sequence turns into a geometric object. In degree $2p$ a Hodge class must have vanishing components of all types $(p + a, p - a)$ with $a \neq 0$. Griffiths transversality then

controls how this condition moves in families, but it produces no object analogous to a line bundle.

Step 3 has no analogue at all

Even on a compact Kähler manifold, Steps 1 and 2 go through, and every integral $(1, 1)$ class is the Chern class of a holomorphic line bundle. It is only Step 3 that uses projectivity, and on a non-algebraic complex torus the line bundle produced in Step 1 may have no meromorphic sections at all. In higher codimension there is not even a line bundle to start from. The nearest substitute is a coherent sheaf, or a vector bundle, with prescribed Chern character. On a very general complex torus of Weil type, Voisin showed that no such sheaf exists [Voi02b]. On an abelian variety we do not know.

These three failures explain the shape of what follows. Since there is no general mechanism for producing cycles, the known results for abelian varieties all start from divisors, the only cycles we can always produce, and move them around with correspondences. Chapter 4 shows how far that takes us.

Chapter 4

Abelian varieties

Abelian varieties are the one large class of varieties whose Hodge classes can be written down. The reason is that their cohomology is an exterior algebra on the first cohomology, so that everything is decided by a single Hodge structure of weight one and the group that preserves it. This chapter sets up that dictionary. The references for the background are Mumford's book [Mum70] and the book of Birkenhake and Lange [BL04]. We prove the facts about Hodge classes, since they are the point of the chapter.

4.1 Complex tori and their cohomology

Let $\mathfrak{t}$ be a complex vector space of dimension g and let $\Lambda \subset \mathfrak{t}$ be a lattice, that is, a discrete subgroup of rank $2g$. The quotient $T = \mathfrak{t}/\Lambda$ is a compact complex manifold and a commutative complex Lie group, and it is called a *complex torus*. Its fundamental group is Λ , and as a real manifold it is a product of $2g$ circles.

Proposition 4.1. *Cup product induces isomorphisms*

$$\bigwedge^k H^1(T, \mathbb{Z}) \xrightarrow{\sim} H^k(T, \mathbb{Z}), \quad H^1(T, \mathbb{Z}) = \text{Hom}(\Lambda, \mathbb{Z}).$$

With complex coefficients, $H^{1,0}(T)$ is the space of $\mathbb{C}$ -linear forms on $\mathfrak{t}$, $H^{0,1}(T)$ is the space of $\mathbb{C}$ -antilinear forms, and

$$H^{p,q}(T) = \bigwedge^p H^{1,0}(T) \otimes \bigwedge^q H^{0,1}(T).$$

Proof. For a product of circles, the Künneth formula shows that the integral cohomology ring is the exterior algebra on the degree-one classes dual to the circles. These classes are a basis of $\text{Hom}(\Lambda, \mathbb{Z})$. For the second part, give T the flat metric coming from a hermitian inner product on $\mathfrak{t}$. A form with constant coefficients in the linear coordinates $z_1, \dots, z_g$ is harmonic, and the constant forms $dz_I \wedge d\bar{z}_J$ with $|I| = p$ and $|J| = q$ span a space of dimension $\binom{g}{p} \binom{g}{q}$. Summing over $p + q = k$ gives $\binom{2g}{k}$, which is the k -th Betti number, so these forms are all the harmonic forms, and their types give the Hodge decomposition. In degree one, dz_j is a $\mathbb{C}$ -linear form and $d\bar{z}_j$ an antilinear one. $\square$

So the Hodge structure of T in every degree is determined by the one in degree one. Conversely, an integral Hodge structure of weight one, of type $\{(1,0), (0,1)\}$, determines a complex torus: one takes the dual of $H^{1,0}$ modulo the lattice H_1 . Complex tori and integral

Hodge structures of this type are the same thing, and homomorphisms of tori are the same as morphisms of these Hodge structures.

4.2 Line bundles and polarisations

By Proposition 4.1, $H^2(T, \mathbb{Z}) = \bigwedge^2 \text{Hom}(\Lambda, \mathbb{Z})$ is the group of alternating forms $E: \Lambda \times \Lambda \rightarrow \mathbb{Z}$. Write $E_{\mathbb{R}}$ for the extension of E to $\mathfrak{t}$, viewed as a real vector space, and i for the complex structure of $\mathfrak{t}$.

Proposition 4.2. *An alternating form E on Λ is of type $(1, 1)$, as a class in $H^2(T, \mathbb{C})$, if and only if $E_{\mathbb{R}}(ix, iy) = E_{\mathbb{R}}(x, y)$ for all $x, y \in \mathfrak{t}$. Every such integral form is the first Chern class of a holomorphic line bundle on T .*

Proof. Extend E complex-bilinearly to $\mathfrak{t} \otimes_{\mathbb{R}} \mathbb{C} = \mathfrak{t}' \oplus \mathfrak{t}''$, the $(+i)$ and $(-i)$ eigenspaces of the complex structure. A class in $\bigwedge^2$ of the dual has type $(1, 1)$ exactly when it vanishes on $\mathfrak{t}' \times \mathfrak{t}'$ and on $\mathfrak{t}'' \times \mathfrak{t}''$. If E is invariant under i , then for $x, y \in \mathfrak{t}'$ we get $E(x, y) = E(ix, iy) = i^2 E(x, y)$, so $E(x, y) = 0$, and the same on $\mathfrak{t}''$. Conversely, if E vanishes on both, write $x = x' + x''$ and $y = y' + y''$. Then

$$E(ix, iy) = E(ix', -iy'') + E(-ix'', iy') = E(x', y'') + E(x'', y') = E(x, y).$$

The second statement is Steps 1 and 2 of the proof of the Lefschetz theorem in Chapter 3, which use only that T is a compact Kähler manifold. $\square$

To such a form E attach the function

$$H(x, y) = E_{\mathbb{R}}(ix, y) + iE_{\mathbb{R}}(x, y), \quad x, y \in \mathfrak{t}.$$

A direct check using the invariance of $E_{\mathbb{R}}$ under i shows that H is a hermitian form, $\mathbb{C}$ -linear in the first variable, with imaginary part E . The Néron–Severi group of T is the group of these forms.

Theorem 4.3 (Riemann, Lefschetz). *The torus T is projective if and only if there is an integral alternating form E , invariant under i , whose hermitian form H is positive definite. In that case E is the Chern class of an ample line bundle.*

One direction is easy. If T embeds in projective space, the restriction of the Fubini–Study form is a positive $(1, 1)$ form, its average under translations has constant coefficients and the same class, and its hermitian form is positive definite. The other direction is the theorem of Lefschetz that the third power of such a line bundle is very ample. It is proved with theta functions [Mum70].

An *abelian variety* is a projective complex torus. A *polarisation* is the class of an ample line bundle, or equivalently a form E as in Theorem 4.3. The form E is a polarisation, in the sense of Chapter 2, of the Hodge structure $H_1(A, \mathbb{Q}) = \Lambda_{\mathbb{Q}}$. Since E is nondegenerate it identifies $\Lambda_{\mathbb{Q}}$ with its dual $H^1(A, \mathbb{Q})$, and we use the same letter for the induced form on $H^1(A, \mathbb{Q})$. From now on we write

$$V = H^1(A, \mathbb{Q}),$$

so that $H^k(A, \mathbb{Q}) = \bigwedge^k V$.

4.3 Endomorphisms and the Rosati involution

A homomorphism of abelian varieties is a holomorphic map that is a group homomorphism. It lifts to a $\mathbb{C}$ -linear map of universal covers that carries one lattice into the other. Write $\text{End}(A)$ for the ring of endomorphisms and $\text{End}^0(A) = \text{End}(A) \otimes \mathbb{Q}$. An element of $\text{End}^0(A)$ acts on $\Lambda_{\mathbb{Q}}$, and its real extension commutes with i . Conversely, a $\mathbb{Q}$ -linear map of $\Lambda_{\mathbb{Q}}$ whose real extension commutes with i has an integer multiple preserving Λ , so it lies in $\text{End}^0(A)$. In other words, $\text{End}^0(A)$ is the algebra of endomorphisms of the Hodge structure $H_1(A, \mathbb{Q})$.

By Poincaré's complete reducibility theorem, every abelian variety is isogenous to a product of powers of simple, pairwise non-isogenous abelian varieties, and $\text{End}^0(A)$ is a semisimple $\mathbb{Q}$ -algebra [Mum70].

Fix a polarisation E . For $\varphi \in \text{End}^0(A)$, acting on $\Lambda_{\mathbb{Q}}$, let φ' be its adjoint with respect to E :

$$E(\varphi x, y) = E(x, \varphi' y).$$

Proposition 4.4. *The adjoint φ' lies in $\text{End}^0(A)$, and the map $\varphi \mapsto \varphi'$ is an involution of $\text{End}^0(A)$, called the Rosati involution. It is positive: $\text{Tr}(\varphi\varphi') > 0$ for $\varphi \neq 0$, the trace being taken on $\Lambda_{\mathbb{Q}}$.*

Proof. Invariance of E under i gives $E(iu, v) = -E(u, iv)$ for all u, v . Using this twice, and that φ commutes with i ,

$$E(x, \varphi' iy) = E(\varphi x, iy) = -E(i\varphi x, y) = -E(\varphi ix, y) = -E(ix, \varphi' y) = E(x, i\varphi' y),$$

so φ' commutes with i and lies in $\text{End}^0(A)$. It is clearly an involution. For positivity, a similar computation shows $H(\varphi x, y) = H(x, \varphi' y)$, so φ' is the adjoint of φ for the positive definite hermitian form H . The trace of $\varphi\varphi'$ on the real space $\mathfrak{t}$ is twice the real part of its complex trace, and the complex trace of $\varphi\varphi^*$ is the sum of the squared lengths of the columns of φ in an H -orthonormal basis, which is positive. $\square$

Proposition 4.5. *The map that sends a symmetric element $\varphi = \varphi'$ of $\text{End}^0(A)$ to the form $E_{\varphi}(x, y) = E(\varphi x, y)$ is an isomorphism*

$$\{\varphi \in \text{End}^0(A) : \varphi' = \varphi\} \xrightarrow{\sim} \text{NS}(A) \otimes \mathbb{Q}.$$

Proof. If φ is symmetric, E_{φ} is alternating, because $E(\varphi y, x) = E(y, \varphi x) = -E(\varphi x, y)$. It is invariant under i , because φ commutes with i and E is invariant. So it lies in $\text{NS}(A)_{\mathbb{Q}}$ by Proposition 4.2. Conversely, given an alternating form E' that is rational and invariant under i , the nondegeneracy of E gives a unique $\mathbb{Q}$ -linear φ with $E'(x, y) = E(\varphi x, y)$. The computation in the proof of Proposition 4.4, applied to E' and E , shows that φ commutes with i , so it lies in $\text{End}^0(A)$, and it is symmetric because E' is alternating. $\square$

So the divisor classes of A are controlled by its endomorphisms. This is the first sign of a principle that runs through the book: whatever is built from divisors is invisible to anything that commutes with the endomorphisms and preserves the polarisation.

The simple case is governed by Albert's classification. If A is simple, $\text{End}^0(A)$ is a division algebra of finite dimension over $\mathbb{Q}$ with a positive involution, and such algebras are of four types: a totally real field; a totally indefinite or a totally definite quaternion algebra over a totally real field; or a division algebra over a CM field, with an involution that restricts

to complex conjugation on the centre [Mum70]. Abelian varieties of Weil type, the subject of Chapter 5, have an endomorphism algebra of the fourth type in its simplest form, an imaginary quadratic field.

4.4 The Hodge group of an abelian variety

Write $\mathrm{Hg}(A)$ for the Hodge group of the Hodge structure $V = H^1(A, \mathbb{Q})$, as defined in Section 2.3.

Proposition 4.6. (i) $\mathrm{Hg}(A)$ is contained in the symplectic group $\mathrm{Sp}(V, E)$.

(ii) The commutant of $\mathrm{Hg}(A)$ in $\mathrm{End}_{\mathbb{Q}}(V)$ is $\mathrm{End}^0(A)$, acting on V by pullback.

(iii) The Hodge classes on A are the invariants of $\mathrm{Hg}(A)$:

$$\mathrm{Hdg}^*(A) = \left(\bigwedge^* V \right)^{\mathrm{Hg}(A)}.$$

(iv) For every $m \geq 1$, the Hodge group of A^m is $\mathrm{Hg}(A)$ acting diagonally on $H^1(A^m, \mathbb{Q}) = V^{\oplus m}$.

Proof. (i) The polarisation is a morphism of Hodge structures $V \otimes V \rightarrow \mathbb{Q}(-1)$. On $\mathbb{Q}(-1)$, of type $(1, 1)$, an element z of the unit circle acts by $z\bar{z} = 1$. So $h(z)$ preserves E for $|z| = 1$. The stabiliser of E is the algebraic group $\mathrm{Sp}(V, E)$, defined over $\mathbb{Q}$, so it contains $\mathrm{Hg}(A)$ by minimality.

(ii) The centraliser of a $\mathbb{Q}$ -linear map φ is an algebraic subgroup defined over $\mathbb{Q}$, so φ commutes with $\mathrm{Hg}(A)$ if and only if it commutes with $h(z)$ for all $|z| = 1$. The element $h(z)$ acts on $V^{1,0}$ and $V^{0,1}$ by different scalars when $z \neq \pm 1$, so commuting with all of them means preserving the Hodge decomposition. That is, φ is an endomorphism of the Hodge structure V , which by the dictionary of the last section means an element of $\mathrm{End}^0(A)$.

(iii) The space $\bigwedge^k V$ is the image of the antisymmetrisation projector on $V^{\otimes k}$, which is both a morphism of Hodge structures and $\mathrm{GL}(V)$ -equivariant. So an element of $\bigwedge^k V$ is a Hodge class, or an invariant, if and only if it is one in $V^{\otimes k}$. Now apply Theorem 2.6.

(iv) The homomorphism defining the Hodge structure of A^m is the diagonal $h \oplus \cdots \oplus h$. The diagonal embedding $\mathrm{GL}(V) \rightarrow \mathrm{GL}(V^{\oplus m})$ is a closed embedding defined over $\mathbb{Q}$, so the smallest $\mathbb{Q}$ -algebraic subgroup containing the diagonal image of $h(U(1))$ is the diagonal image of $\mathrm{Hg}(A)$. $\square$

Part (iv) means that the representation of one reductive group on one vector space decides the Hodge classes not only on A but on all its powers, and hence on every abelian variety isogenous to a product of copies of A . Part (iii) turns the computation of Hodge classes into invariant theory.

For a very general principally polarised abelian variety the Hodge group is the whole symplectic group. One way to see this is through monodromy. Over the moduli space with level structure, the monodromy of H^1 is a subgroup of finite index in $\mathrm{Sp}_{2g}(\mathbb{Z})$, which is Zariski dense in Sp_{2g} . By André's theorem [And92], the algebraic monodromy group is contained in the Hodge group of a very general member. So that Hodge group contains Sp_{2g} , and by part (i) it equals it.

4.5 The Lefschetz group and Lefschetz classes

Definition 4.7. The *Lefschetz group* $L(A)$ is the connected component of the identity of the centraliser of $\text{End}^0(A)$ in $\text{Sp}(V, E)$. The *Lefschetz classes* of A are the elements of the subalgebra $\mathcal{D}^*(A) \subseteq H^*(A, \mathbb{Q})$ generated by divisor classes.

By Proposition 4.6(i) and (ii), and because $\text{Hg}(A)$ is connected,

$$\text{Hg}(A) \subseteq L(A).$$

The Lefschetz group is easy to compute, since it depends only on the endomorphism algebra with its involution. The Hodge group is hard to compute, since it depends on the period point.

Proposition 4.8. *Every divisor class on A is invariant under $L(A)$. Hence*

$$\mathcal{D}^*(A) \subseteq \left(\bigwedge^* V \right)^{L(A)} \subseteq \left(\bigwedge^* V \right)^{\text{Hg}(A)} = \text{Hdg}^*(A).$$

Proof. It is enough to work on H_1 , where $L(A)$ acts through the contragredient. By Proposition 4.5, a divisor class is a form $E(\varphi x, y)$ with φ in $\text{End}^0(A)$. An element g of $L(A)$ preserves E and commutes with φ , so $E(\varphi g x, g y) = E(g \varphi x, g y) = E(\varphi x, y)$. The middle inclusion holds because $\text{Hg}(A) \subseteq L(A)$, and the last equality is Proposition 4.6(iii). $\square$

The first inclusion is in fact an equality. This theorem is due to Murty and Milne [Mur84, Mil99]: the Lefschetz classes are exactly the invariants of the Lefschetz group. We shall use only the inclusion proved above, together with the following consequence of the theorem.

Corollary 4.9. *If $\text{Hg}(A)$ and $L(A)$ have the same invariants in $\bigwedge^* V$, and in particular if $\text{Hg}(A) = L(A)$, then every Hodge class on A is a polynomial in divisor classes, and the Hodge conjecture holds for A .*

This covers a great deal. For a very general principally polarised abelian variety, both groups are the symplectic group, and the Hodge ring is generated by the class of the theta divisor. Many classes of abelian varieties are known to satisfy the hypothesis of the corollary, among them simple abelian varieties of prime dimension and products of elliptic curves [Rib83, BL04]. Moonen and Zarhin worked through all abelian varieties of dimension at most five, and in that range the Hodge classes that are not Lefschetz classes are all accounted for by Weil classes [MZ99].

The first place where the hypothesis fails is the subject of the next chapter. There the endomorphism algebra is an imaginary quadratic field K , the Lefschetz group is a unitary group, and the Hodge group is the special unitary group, one dimension smaller. The difference is a single character, the determinant, and it is exactly enough to create Hodge classes that no divisor can explain.

Chapter 5

Abelian varieties of Weil type

In 1977 Weil wrote a short paper with a simple observation [Wei77]. Let an imaginary quadratic field act on an abelian variety so that its two eigenspaces on the tangent space have the same dimension. Then the determinant of the first cohomology over that field gives a plane of Hodge classes, and for a general such variety these classes are not products of divisor classes. This chapter proves Weil's observation, describes the family in which these varieties live, computes the Hodge ring of its general member, and shows that the Hodge conjecture for that member is the algebraicity of one class. The last section looks at the same classes on complex tori that are not algebraic, where Voisin showed that they are not algebraic in any sense.

5.1 Definition

Fix an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$, with d a positive squarefree integer, and write $\tau \mapsto \bar{\tau}$ for its nontrivial automorphism. Fix one embedding of K in $\mathbb{C}$.

Definition 5.1. An abelian variety A of dimension $2n$ with a polarisation E is *of Weil type* for K if it carries an embedding $K \hookrightarrow \text{End}^0(A)$ such that

- (i) the Rosati involution of E restricts to complex conjugation on K , and
- (ii) the element $\sqrt{-d}$ acts on the tangent space at the origin with the eigenvalues $\sqrt{-d}$ and $-\sqrt{-d}$ each of multiplicity n .

The second condition is the one that matters. Without it the construction below still makes sense, but the classes it produces are not of Hodge type.

Let $V = H^1(A, \mathbb{Q})$. Since K is commutative, pullback makes V a K -vector space, of dimension $2n$ because $\dim_{\mathbb{Q}} V = 4n$. Over $\mathbb{C}$ the action of $\sqrt{-d}$ splits $V_{\mathbb{C}}$ into two eigenspaces

$$V_{\mathbb{C}} = V_+ \oplus V_-, \quad \dim V_{\pm} = 2n,$$

on which $\sqrt{-d}$ acts by $\pm\sqrt{-d}$. Complex conjugation of coefficients exchanges them. Because K acts by endomorphisms, it preserves the Hodge decomposition, so each eigenspace splits as $V_{\pm} = V_{\pm}^{1,0} \oplus V_{\pm}^{0,1}$. The space $V^{1,0}$ is dual to the tangent space, so condition (ii) says that $V_+^{1,0}$ and $V_-^{1,0}$ both have dimension n . Since $V_+^{0,1} = \overline{V_-^{1,0}}$, we get

$$\dim V_+^{1,0} = \dim V_+^{0,1} = n, \quad \dim V_-^{1,0} = \dim V_-^{0,1} = n. \quad (5.1)$$

The hermitian form

Condition (i) says that $E(\tau x, y) = E(x, \bar{\tau} y)$ for $\tau \in K$. Fix $\delta \in K$ with $\bar{\delta} = -\delta$, for instance $\delta = \sqrt{-d}$. Then there is a unique K -hermitian form $H: V \times V \rightarrow K$, linear in the first variable, with

$$E(x, y) = \text{Tr}_{K/\mathbb{Q}}(\delta H(x, y)).$$

Indeed, for fixed y the map $x \mapsto E(x, y)$ is $\mathbb{Q}$ -linear, and the trace form identifies $\mathbb{Q}$ -linear forms on V with K -linear forms; the compatibility with K makes the resulting form hermitian. Its determinant, in a K -basis, is well defined in $\mathbb{Q}^\times / N_{K/\mathbb{Q}}(K^\times)$, and it is called the *discriminant*. The unitary group $U(V, H)$ is the group of K -linear automorphisms of V that preserve E . Its subgroup of elements of determinant one over K is the special unitary group $SU(V, H)$. Both are algebraic groups over $\mathbb{Q}$.

Lemma 5.2. *The eigenspaces V_+ and V_- are isotropic for E , and E pairs them perfectly. So, as representations of $U(V, H)_\mathbb{C} \cong \text{GL}(V_+)$, the space V_- is the dual of V_+ .*

Proof. For $x, y \in V_+$ we have $E(\sqrt{-d}x, \sqrt{-d}y) = E(x, \overline{\sqrt{-d}}\sqrt{-d}y) = dE(x, y)$, while bilinearity gives $E(\sqrt{-d}x, \sqrt{-d}y) = -dE(x, y)$. So $E(x, y) = 0$, and the same holds on V_- . Since E is nondegenerate on $V_\mathbb{C} = V_+ \oplus V_-$ and vanishes on each summand, it pairs them perfectly. An element of $U(V, H)$ commutes with K , so it preserves V_+ and V_- , and it is determined by its restriction to V_+ ; it preserves E , so its action on V_- is the contragredient. $\square$

The form $h(x, y) = \delta E(x, \bar{y})$, for $x, y \in V_+$, is hermitian on V_+ . It is the complexification of H along the fixed embedding of K , up to a positive constant. The Hodge–Riemann relations say that h is definite of one sign on $V_+^{1,0}$ and of the other on $V_+^{0,1}$, so by (5.1) it has signature (n, n) . This is the form in which condition (ii) is usually stated: the hermitian form of an abelian variety of Weil type has signature (n, n) .

5.2 The Weil classes

Definition 5.3. The *Weil plane* of A is

$$W = W(A) = \left(\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_- \right) \cap \bigwedge_{\mathbb{Q}}^{2n} V \subseteq H^{2n}(A, \mathbb{Q}).$$

Its elements are the *Weil classes*.

The two lines $\bigwedge^{2n} V_\pm$ sit inside $\bigwedge^{2n} V_\mathbb{C} = \bigoplus_{a+b=2n} \bigwedge^a V_+ \otimes \bigwedge^b V_-$ as the summands with $(a, b) = (2n, 0)$ and $(0, 2n)$. Complex conjugation exchanges them, so their sum is defined over $\mathbb{Q}$, and W is a rational plane. The field K acts on W : an element τ acts on $\bigwedge^{2n} V_+$ by τ^{2n} and on $\bigwedge^{2n} V_-$ by $\bar{\tau}^{2n}$. So W is a one-dimensional K -vector space, canonically the K -determinant $\bigwedge_K^{2n} V$.

Theorem 5.4 (Weil). *The Weil classes are Hodge classes.*

Proof. By (5.1),

$$\bigwedge^{2n} V_+ = \bigwedge^n V_+^{1,0} \otimes \bigwedge^n V_+^{0,1},$$

which is of type (n, n) . The same holds for V_- . $\square$

The proof shows exactly why the balanced signature is needed. If the eigenspace V_+ had a holomorphic and b antiholomorphic dimensions with $a \neq b$, its top exterior power would have type (a, b) .

There is a second proof, through groups, which explains more. The Hodge group of A commutes with K and preserves E , so it lies in $U(V, H)$. On V_+ an element z of the unit circle acts through $h(z)$ by z^{-1} on $V_+^{1,0}$ and by z on $V_+^{0,1}$, so its determinant on V_+ is $z^{-n}z^n = 1$. Thus $h(U(1))$ lies in the special unitary group, and by minimality

$$\mathrm{Hg}(A) \subseteq \mathrm{SU}(V, H). \quad (5.2)$$

The special unitary group acts trivially on $\bigwedge^{2n} V_\pm$, since it acts there by the determinant. So the Weil classes are invariants of $\mathrm{Hg}(A)$, which is another way of saying they are Hodge classes.

The same computation shows why they are *not* Lefschetz classes in general. The centre of $U(V, H)$ is the torus $U(1)_K$ of norm-one elements of K , and $\lambda \in U(1)_K$ acts on $\bigwedge^{2n} V_+$ by λ^{2n} .

Proposition 5.5. *Suppose $\mathrm{End}^0(A) = K$. Then the Lefschetz group of A is $U(V, H)$, and no nonzero Weil class is a Lefschetz class.*

Proof. The centraliser of K in $\mathrm{Sp}(V, E)$ consists of the K -linear maps that preserve E , which is $U(V, H)$, a connected group. So $L(A) = U(V, H)$. Choose $\tau \in K^\times$ such that $\lambda = \tau/\bar{\tau}$ is not a root of unity. This is possible because K contains finitely many roots of unity ζ , and each equation $\tau = \zeta\bar{\tau}$ cuts out a line in the plane K . Then λ has norm one. Then $\lambda^{2n} \neq 1$ and $\bar{\lambda}^{2n} \neq 1$, so λ moves every nonzero vector of W . By Proposition 4.8, Lefschetz classes are invariant under $L(A)$, so they meet W only in 0. $\square$

This is Weil's theorem: on an abelian variety of Weil type whose endomorphism algebra is no bigger than K , the Weil classes are Hodge classes that no polynomial in divisor classes can produce. When $n = 1$ the hypothesis never holds. The Weil classes then lie in H^2 , so they are divisor classes by the theorem of Lefschetz, and the proposition shows that the endomorphism algebra of an abelian surface of Weil type is always larger than K .

5.3 The family

All abelian varieties of Weil type for K with a given hermitian space (V, H) fit into one family. We describe its base.

Fix (V, H) of signature (n, n) , and let h be the hermitian form on V_+ as above. A polarised Hodge structure of weight one on V , of Weil type for K , is determined by the n -dimensional subspace $X = V_+^{1,0} \subset V_+$. Indeed, $V_+^{1,0}$ is isotropic for E by the Hodge–Riemann relations, so $V_-^{1,0}$ is the annihilator of X in V_- under the pairing of Lemma 5.2, and then $V_-^{0,1}$ is the complex conjugate of $V_+^{1,0}$. The positivity part of the Hodge–Riemann relations says that h , normalised with the right sign, is positive definite on X . Conversely, every n -dimensional subspace on which h is positive definite arises in this way. So the parameter space is

$$D = \{X \subset V_+ : \dim X = n, h|_X > 0\},$$

an open subset of the Grassmannian of n -planes in a space of dimension $2n$. It has dimension n^2 , and it is the bounded symmetric domain of the real group $\mathrm{SU}(n, n)$.

Let Γ be a torsion-free arithmetic subgroup of $\mathrm{SU}(V, H)(\mathbb{Q})$ that preserves a lattice in V . The quotient

$$\mathrm{Sh} = \Gamma \backslash D$$

is a smooth quasi-projective variety by the theorem of Baily and Borel [BB66]. Over it there is a polarised abelian scheme $\pi: \mathcal{A} \rightarrow \mathrm{Sh}$ whose fibre over the class of X is the abelian variety with $H^1 = V$ and $V_+^{1,0} = X$. We call it the *Weil family*. Its monodromy group on H^1 is Γ .

Proposition 5.6. *The Weil plane is a constant local subsystem of $R^{2n}\pi_*\mathbb{Q}$, and every Weil class extends to a global flat section that is a Hodge class at every point.*

Proof. The monodromy group Γ lies in $\mathrm{SU}(V, H)$, which acts trivially on W . So the monodromy fixes W pointwise. By Theorem 5.4, applied at each point, the transported classes are Hodge classes everywhere. $\square$

Proposition 5.7. *Let $n \geq 2$. A very general member A of the Weil family has $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ and $\mathrm{End}^0(A) = K$. The set of members where the Hodge group is smaller is a countable union of proper closed algebraic subsets of Sh .*

Proof. The real group $\mathrm{SU}(n, n)$ has no compact factor, and $\mathrm{SU}(V, H)$ is simple over $\mathbb{Q}$, so by the Borel density theorem the arithmetic subgroup Γ is Zariski dense in $\mathrm{SU}(V, H)$. By André's theorem [And92], the connected algebraic monodromy group is contained in the Hodge group of a very general member. So that Hodge group contains $\mathrm{SU}(V, H)$, and by (5.2) it equals it. For the endomorphisms, over $\mathbb{C}$ the group $\mathrm{SU}(V, H)$ is $\mathrm{SL}(V_+)$, acting on V_+ by the standard representation and on V_- by its dual (Lemma 5.2). For $2n \geq 3$ these are irreducible and not isomorphic, so the commutant of $\mathrm{SU}(V, H)$ in $\mathrm{End}_{\mathbb{Q}}(V)$ has dimension two, and it contains K . By Proposition 4.6(ii) it is $\mathrm{End}^0(A)$. For the last statement, the members with a smaller Hodge group are those carrying an extra Hodge class in some tensor space. Each such class has an algebraic Hodge locus by Theorem 2.11, there are countably many rational classes, and none of these loci is all of Sh . $\square$

We write Sh° for the set of points where the Hodge group is all of $\mathrm{SU}(V, H)$. Its complement is meagre.

5.4 The Hodge ring of the general member

We can now compute every Hodge class on a member with full Hodge group. The answer is as small as it could be, given Weil's theorem.

Theorem 5.8. *Let $n \geq 2$ and let A be a member of the Weil family with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$. Then*

$$\mathrm{Hdg}^p(A) = \mathbb{Q}\eta^p \quad (p \neq n), \quad \mathrm{Hdg}^n(A) = \mathbb{Q}\eta^n \oplus W,$$

where η is the class of the polarisation.

Proof. By Proposition 4.6(iii) we need the invariants of $\mathrm{SU}(V, H)$ in $\wedge^* V$. Over $\mathbb{C}$,

$$\wedge^k V_{\mathbb{C}} = \bigoplus_{a+b=k} \wedge^a V_+ \otimes \wedge^b V_-,$$

and as a representation of $\mathrm{SL}(V_+)$ the summand is $\wedge^a S \otimes (\wedge^b S)^\vee$, where $S = V_+$ is the standard representation. Its invariants are the $\mathrm{SL}(V_+)$ -homomorphisms $\wedge^b S \rightarrow \wedge^a S$. The

exterior powers $\bigwedge^a S$, $0 \leq a \leq 2n$, are irreducible, and two of them are isomorphic only when $a = b$, or when both are trivial, which happens for $a, b \in \{0, 2n\}$. So there is a one-dimensional space of invariants in each summand with $a = b$, and in the two summands $(a, b) = (2n, 0)$ and $(0, 2n)$, and none elsewhere.

The summands with $(a, b) = (2n, 0)$ and $(0, 2n)$ make up the Weil plane. In the summand with $a = b = p$, the class η^p is a nonzero invariant for $0 \leq p \leq 2n$. Indeed η is invariant, and $\eta^p \neq 0$ because η^{2n} is a positive multiple of the volume class. Its component in that summand is nonzero, because the other summands of $\bigwedge^{2p} V$ contain no invariants except the Weil plane when $p = n$, and η^n is not a Weil class: the centre $U(1)_K$ fixes η^n and moves every nonzero Weil class. So η^p spans the invariants with $a = b = p$. $\square$

Corollary 5.9. *Let $n \geq 2$ and $A \in \text{Sh}^\circ$. The following are equivalent.*

- (i) *The Hodge conjecture holds for A in every degree.*
- (ii) *Some nonzero Weil class on A is algebraic.*
- (iii) *Every Weil class on A is algebraic.*

Proof. By Theorem 5.8, (i) and (iii) are equivalent, since the powers of η are algebraic. Suppose a nonzero Weil class ω is the class of a cycle Z . Choose $\tau \in K$ with $\tau^{2n} \notin \mathbb{Q}$, for example $\tau = a + \sqrt{-d}$ with a a positive integer chosen so that $\tau/\bar{\tau}$ is not a root of unity. Some positive integer multiple $m\tau$ is an endomorphism of A , and its pullback acts on W by $(m\tau)^{2n}$. So $(m\tau)^*Z$ is a cycle with class $m^{2n}\tau^{2n}\omega$. The classes ω and $\tau^{2n}\omega$ are linearly independent over $\mathbb{Q}$, since W is a one-dimensional K -space and $\tau^{2n} \notin \mathbb{Q}$. So they span W , and every Weil class is algebraic. $\square$

For a very general abelian variety of Weil type, then, the whole Hodge conjecture is a statement about one class. That is the sense in which these varieties are the first test case. They are also, by Proposition 5.5, exactly the case that divisors cannot reach.

5.5 Weil tori

Drop the polarisation, and keep only the action of K with balanced signature. Let V be as before. For complementary n -dimensional subspaces $X, Y \subset V_+$, the subspace $X \oplus \bar{Y}$ of $V_{\mathbb{C}}$ is the space of holomorphic one-forms of a complex structure on the real torus $V_{\mathbb{R}}^{\vee}/H_1$. Call the resulting complex torus $T_{X,Y}$. On it K acts with balanced signature, and every complex structure commuting with K with balanced signature arises in this way. The pairs (X, Y) form a connected complex manifold $\mathcal{S}$ of dimension $2n^2$, and the Weil family is the submanifold of dimension n^2 where Y is the h -orthogonal complement of X and h is positive on X . Since $\bigwedge^{2n} V_+ = \bigwedge^n X \otimes \bigwedge^n Y$, with X holomorphic and Y antiholomorphic, the Weil classes are of type (n, n) on every $T_{X,Y}$. These are the *Weil tori*.

Proposition 5.10. *For a very general Weil torus T , the Hodge classes are W in degree $2n$ and zero in every other degree strictly between 0 and $4n$.*

Proof. There are countably many rational classes, and the Hodge locus of each is an analytic subset of $\mathcal{S}$. So a class that is a Hodge class at a very general point is a Hodge class at every point. Let $G = \text{Res}_{K/\mathbb{Q}} \text{GL}_K(V)$ be the group of K -linear automorphisms of V . Its real points,

which form the group $\mathrm{GL}(V_+)$, act transitively on pairs of complementary subspaces of V_+ , hence on $\mathcal{S}$. The Mumford–Tate group M of a very general member lies in G , because K acts by endomorphisms of every member, and it contains the Mumford–Tate group of every member. So the real Lie algebra of M contains the complex structure of every member, and hence the span of the orbit of one of them under the adjoint action of $G(\mathbb{R})$. That span is an ideal of $\mathfrak{gl}(V_+)$. It contains a non-central element, the complex structure, which acts by i on X and by $-i$ on Y . So it contains $\mathfrak{sl}(V_+)$, and M contains the derived group of G . Over $\mathbb{C}$ that derived group is $\mathrm{SL}(V_+) \times \mathrm{SL}(V_-)$, acting independently on the two factors of $\Lambda^* V_+ \otimes \Lambda^* V_-$. Its invariants are spanned by 1 , $\Lambda^{2n} V_+$, $\Lambda^{2n} V_-$ and their product. Hodge classes are invariant under M , which proves the statement. $\square$

Notice the difference from the polarised case. There the two factors are dual to each other under one group, and the classes η^p appear. On a very general Weil torus there is no polarisation, and nothing but the Weil plane survives.

Lemma 5.11. *A very general Weil torus has no analytic subsets other than finite sets and itself.*

Proof. Let Z be an irreducible analytic subset of codimension c with $0 < c < 2n$. For a Kähler class κ we have $\int_T [Z] \cup \kappa^{2n-c} = \int_Z \kappa^{2n-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$. By Proposition 5.10, $c = n$ and $[Z] \in W$. Take for κ_0 the class of a constant positive definite hermitian form that is block diagonal for the splitting $H^{1,0} = X \oplus \bar{Y}$. Its two blocks lie in $X \otimes \bar{X}$ and $\bar{Y} \otimes Y$, both inside $V_+ \otimes V_-$. So every term of κ_0^n has n factors from V_+ , and its product with an element of $\Lambda^{2n} V_+$ vanishes, since $\Lambda^{3n} V_+ = 0$. The same holds with V_- . Hence $\int_T [Z] \cup \kappa_0^n = 0$, a contradiction. $\square$

Voisin proved much more. Coherent sheaves are the natural generalisation of subvarieties on a complex manifold that need not be projective, and one can ask whether the Weil classes are at least combinations of Chern classes of coherent sheaves. They are not.

Theorem 5.12 (Voisin [Voi02b]). *Let $n = 2$ and let T be a very general Weil torus. Every coherent analytic sheaf on T has $c_2 = 0$ in $H^4(T, \mathbb{Q})$. In particular the Weil classes of T are Hodge classes that are not combinations of Chern classes of coherent sheaves.*

We sketch the proof, because the same argument works for every $n \geq 2$ and it is used in Chapter 7. Chern classes of coherent sheaves on a compact complex manifold exist in rational Deligne cohomology [Gri10], and they are Hodge classes. By Lemma 5.11, the torsion of a coherent sheaf and the locus where a torsion-free sheaf fails to be reflexive are supported on finite sets, and they contribute only in top degree. So it is enough to treat reflexive sheaves. Since there are no Hodge classes in degree two, every sheaf has $c_1 = 0$, so every torsion-free sheaf has slope zero for every Kähler class. Stability then does not depend on the Kähler class, and a Jordan–Hölder filtration reduces the question to stable reflexive sheaves Q . For these, the Bogomolov inequality, in the form of Bando and Siu for reflexive sheaves [BS94], gives

$$\int_T c_2(Q) \cup \kappa^{2n-2} \geq 0$$

for every Kähler class κ , with equality only if Q is a flat vector bundle. For $n \geq 3$ the class $c_2(Q)$ is a Hodge class of degree four, hence zero. For $n = 2$ it lies in W , and the class κ_0 of

the proof of Lemma 5.11 gives equality. Either way Q is flat, and its Chern classes vanish. The same argument shows that every coherent sheaf on a very general Weil torus of dimension $2n$ has $\text{ch}_k = 0$ for $0 < k < 2n$.

Voisin's theorem is a counterexample to the Hodge conjecture for compact Kähler manifolds, as explained in Chapter 1. For us it plays a different role. It is a warning about deformation arguments. The Weil class is of Hodge type on every Weil torus, polarised or not. An argument that moves a cycle along every deformation preserving the Hodge type of the Weil class would move it onto a Weil torus, where no cycle or sheaf represents the class. So any such argument has to use the polarisation in an essential way. This is the source of the main obstruction in Chapter 7.

Chapter 6

Where the Weil class is known to be algebraic

This chapter collects the cases in which the Weil classes are known to be algebraic. They fall into three groups. At special members of a Weil family, where the endomorphism algebra is larger than K , the Weil classes can be products of divisor classes, and we prove this in an explicit example. For the field $\mathbb{Q}(\sqrt{-3})$, and for some families for $\mathbb{Q}(\sqrt{-1})$, curves with an automorphism give cycles on Prym varieties, by work of Schoen and Koike. And in dimension four, Markman's secant sheaves settle every family. The last section reports, separately and conditionally, what recent unrefereed preprints announce.

We use the notation of Chapter 5. In particular Sh is the base of a Weil family of dimension $2n$, $\Sigma \subseteq \text{Sh}$ is the set of members on which the Weil classes are algebraic, and by Corollary 5.9 it does not matter whether we ask for one nonzero Weil class or for all of them.

6.1 A member where divisors suffice

Let E be an elliptic curve with complex multiplication by K , so that $\text{End}^0(E) = K$. Let z be a linear coordinate on its universal cover, and normalise the action so that $\tau \in K$ acts on the holomorphic form dz by τ , through the fixed embedding $K \subset \mathbb{C}$. Then τ acts on $d\bar{z}$ by $\bar{\tau}$.

On $A = E^{2n}$ let $\tau \in K$ act by τ on the first n factors and by $\bar{\tau}$ on the last n . Give A the product polarisation. The Rosati involution of E is complex conjugation on K , so the Rosati involution of A restricts to complex conjugation on this copy of K .

Proposition 6.1. *With this action A is of Weil type for K , and its Weil classes are polynomials in divisor classes. In particular they are algebraic.*

Proof. Write $z_1, \dots, z_{2n}$ for the coordinates on the factors. The element τ acts by τ on dz_i for $i \leq n$, and on $d\bar{z}_i$ for $i > n$, because on those factors it acts through $\bar{\tau}$, whose conjugate is τ . So

$$V_+ = \langle dz_1, \dots, dz_n, d\bar{z}_{n+1}, \dots, d\bar{z}_{2n} \rangle,$$

which has n holomorphic and n antiholomorphic directions, and A is of Weil type. Up to sign,

$$\bigwedge^{2n} V_+ = \prod_{i=1}^n (dz_i \wedge d\bar{z}_{n+i}).$$

Each factor $dz_i \wedge d\bar{z}_{n+i}$ lies in the $(1, 1)$ part of $H^1(E_i) \otimes H^1(E_{n+i})$, the Künneth component of $H^2(E \times E)$ that corresponds to $\text{Hom}(E, E)$. That $(1, 1)$ part has complex dimension two, spanned by $dz_i \wedge d\bar{z}_{n+i}$ and $d\bar{z}_i \wedge dz_{n+i}$. On the other hand, the classes of the graphs of the endomorphisms 1 and $\sqrt{-d}$ of E , minus their components in $H^0 \otimes H^2$ and $H^2 \otimes H^0$, give two linearly independent rational classes in that component. So the whole $(1, 1)$ part is spanned over $\mathbb{C}$ by divisor classes, and $\bigwedge^{2n} V_+$ lies in the complex span of products of n divisor classes. The same holds for $\bigwedge^{2n} V_-$, its complex conjugate. So $W \otimes \mathbb{C}$ lies in $\mathcal{D}^n(A) \otimes \mathbb{C}$. Both W and $\mathcal{D}^n(A)$ are rational subspaces, so $W \subseteq \mathcal{D}^n(A)$. $\square$

The example has one discriminant, determined by the product form. A construction with quaternion algebras gives such a member in every Weil family, of every dimension and every discriminant, with the Weil classes again polynomials in divisor classes [BhH8]. We record the consequence and prove the part we need.

Proposition 6.2. *Suppose Σ contains one point of Sh . Then Σ is dense in Sh .*

Proof. Let $s_0 \in \Sigma$ lift to $\tilde{s}_0 \in D$. An element g of $\text{SU}(V, H)(\mathbb{Q})$ is a K -linear automorphism of V preserving E . It is an isomorphism of rational Hodge structures from the structure at $\tilde{s}_0$ to the structure at $g\tilde{s}_0$, so a positive integer multiple of it is an isogeny between the corresponding abelian varieties. The isogeny commutes with K , so it carries Weil classes to Weil classes, and it carries cycles to cycles. So the image of $g\tilde{s}_0$ lies in Σ . It remains to see that the orbit $\text{SU}(V, H)(\mathbb{Q}) \cdot \tilde{s}_0$ is dense in D . The real group $\text{SU}(V, H)(\mathbb{R}) = \text{SU}(n, n)$ acts transitively on D , so it suffices that $\text{SU}(V, H)(\mathbb{Q})$ is dense in $\text{SU}(V, H)(\mathbb{R})$. This is real approximation for the connected group $\text{SU}(V, H)$ [PR94, Chapter 7]. $\square$

So the algebraic locus is never small in the topological sense. What is not known is whether it has interior. Every point that this proof produces is isogenous to s_0 , so it has the same Hodge group as s_0 , and in the example above that group is a torus. Chapter 7 explains why this is the heart of the problem.

6.2 Lowering the dimension

The next statement, due to Schoen, lets a result in one dimension imply a result one step lower.

Proposition 6.3 (Schoen [Sch98]). *Let A be of Weil type for K , of dimension $2n$ and discriminant δ , and let B be an abelian surface of Weil type for K with the same discriminant. Then $A \times B$ is of Weil type for K , of dimension $2n + 2$, and its hermitian form is split, that is, it has a totally isotropic K -subspace of dimension $n + 1$. If the Weil classes of $A \times B$ are algebraic, so are those of A .*

The first part is a computation with hermitian forms. The form of $A \times B$ is the orthogonal sum of the two forms. With the discriminants normalised as in [BhH8], matching discriminants make the sum have the discriminant of a sum of $n+1$ hyperbolic planes. By Landherr's theorem, a hermitian form over K is determined by its rank, its discriminant and its signature, so the sum is hyperbolic. For the second part, the Weil plane of $A \times B$ is the K -tensor product of the Weil planes of A and B . The Weil classes of B lie in H^2 , where they are divisor classes,

and intersecting with a suitable divisor class pulled back from B and pushing forward to A returns a nonzero Weil class of A . A complete proof in coordinates is in [BhH8].

The consequence is that the *split* families in dimension $2n + 2$ control all families in dimension $2n$. This is how Markman passes from sixfolds to fourfolds.

6.3 Curves with an automorphism

Schoen found the first cycles representing Weil classes at members with full Hodge group. Let $\tilde{C} \rightarrow C$ be an étale cyclic cover of degree three of a smooth projective curve of genus $g \geq 2$, with covering automorphism ρ . The Prym variety P is the abelian subvariety of $J(\tilde{C})$ on whose first cohomology ρ has no fixed vector. Then $K = \mathbb{Q}(\sqrt{-3})$ acts on P through ρ . By the Chevalley–Weil formula, the two nontrivial eigenspaces of ρ on holomorphic differentials of $\tilde{C}$ both have dimension $g - 1$, so P is of Weil type with $n = g - 1$. Its hermitian form is split, by a computation of Looijenga with the homology of the cover [Loo97].

Theorem 6.4 (Schoen [Sch88]). *The Weil classes of these Prym varieties are algebraic.*

Schoen’s cycles come from the canonical system of the base curve. On the symmetric power $C^{(2g-2)}$, the canonical system $|K_C| \cong \mathbb{P}^{g-1}$ is the fibre of the Abel–Jacobi map over the canonical class. The cover $\tilde{C} \rightarrow C$ defines an étale cover of $C^{(2g-2)}$ of degree three, and over $|K_C|$, which is simply connected, it splits into three copies of $\mathbb{P}^{g-1}$. The differences of the copies, weighted by the two characters of order three, are cycles on a quotient of $\tilde{C}^{2g-2}$ whose classes span the Weil classes there, and a correspondence through $J(\tilde{C})$ carries them to P [Sch88, Theorem 2.0 with $r = 0$, Corollary 3.1]. Patel and Zhang have recently given a second proof using unramified class field theory (preprint) [PZ25]. Looijenga showed that the monodromy of the family of these Prym varieties is an arithmetic subgroup of the unitary group [Loo97]. By Borel density and André’s theorem, as in Proposition 5.7, a very general Prym of this kind has full Hodge group $\mathrm{SU}(V, H)$. So Schoen’s cycles live at members with full Hodge group. We follow the account in [BhH8].

The family of Pryms has dimension $3g - 3 = 3n$, the dimension of the moduli of curves of genus g . The Weil family has dimension n^2 . For $n = 2$ Schoen showed that the Pryms fill the Weil family of fourfolds of their polarization type, and proved the conjecture there [Sch88, Theorem 3.2]; his addendum points out that the theorem should have carried this hypothesis on the polarization [Sch98]. For $n = 3$ the addendum shows, with a calculation of Faber, that the Pryms fill a family of sixfolds, and then descends through Proposition 6.3 to every Weil fourfold for $\mathbb{Q}(\sqrt{-3})$, whatever its polarization invariant [Sch98]. For $n \geq 4$ they fill a locus of dimension $3n < n^2$. Koike proved the analogous statement for a family of sixfolds for $\mathbb{Q}(\sqrt{-1})$ [Koi04].

In Chapter 7 we prove that cycles built from a curve with an automorphism of order three in the obvious ways, through diagonals, graphs of the automorphism and the group law, never represent a Weil class (Proposition 7.11). Schoen’s cycles therefore use something beyond these constructions, and that is why they reach classes that divisors cannot.

6.4 Secant sheaves

The most complete results are Markman’s. He works with abelian varieties of Weil type of the form $B \times \hat{B}$, for an abelian variety B of dimension n and its dual, with an action of K that

does not respect the product. These are special members of their Weil families. On them he constructs sheaves from the secant varieties of B in a projective embedding, the *secant sheaves*, whose Chern characters have a nonzero Weil component. He then shows that these sheaves deform along the whole family, by an equivariant form of the semiregularity theorem of Bloch, Buchweitz–Flenner and Pridham [Blo72, BF03, Pri24].

Theorem 6.5 (Markman [Mar25a, Mar25b], preprints). *(i) The Weil classes are algebraic on every abelian sixfold of Weil type whose hermitian form is split.*

(ii) The Weil classes are algebraic on every abelian fourfold of Weil type, for every imaginary quadratic field and every discriminant.

Part (ii) follows from part (i) by Proposition 6.3. Combined with the analysis of Moonen and Zarhin [MZ99], it gives the Hodge conjecture for every abelian variety of dimension at most five. At the time of writing, Markman’s papers are on the arXiv, and we mark the results as preprint results accordingly.

It is worth seeing where secant sheaves sit with respect to the barrier of Chapter 7. On $B \times \widehat{B}$ the endomorphism algebra is much larger than K , the Lefschetz group is much smaller than $U(V, H)$, and the Weil classes are invariant under it. So nothing prevents a sheaf built from line bundles from having a Weil component there. The difficulty is moved into the deformation step, which is where all the work lies.

6.5 Recent preprints

In September and October 2026, OpenAI released a series of preprints announcing further cases [OAI26d, OAI26e, OAI26g]. They have not been refereed, and this book does not check them. Nothing outside this section depends on them. The announced statements include:

- (a) the Hodge conjecture for every abelian variety of CM type [OAI26e];
- (b) the algebraicity of the Weil classes on every abelian eightfold of Weil type whose hermitian form is split, for every imaginary quadratic field [OAI26d];
- (c) the Hodge conjecture for every power of an abelian sixfold of split Weil type, and of an abelian variety of dimension at most five with an action of an imaginary quadratic field [OAI26g].

If (b) holds, Proposition 6.3 gives the Weil classes on every abelian sixfold of Weil type [BhH8]. Even then, the Weil classes in dimension eight with non-split form, and in every dimension from ten on, would remain open, and so would the classes on abelian varieties that are not of Weil type and are not Lefschetz, which [BhH8] lists.

6.6 Summary

Dimension $2n$	Status of the Weil classes	Source
4	algebraic, all K and discriminants	Markman (preprint); earlier cases Schoen
6, split form	algebraic	Markman (preprint)
6, $\mathbb{Q}(\sqrt{-3})$, Prym family	algebraic	Schoen
6, one family for $\mathbb{Q}(\sqrt{-1})$	algebraic	Koike
6, other	open; announced via (b)	preprints
8, split form	open; announced	preprint (b)
$2n \geq 8$, $\mathbb{Q}(\sqrt{-3})$	algebraic on a locus of dimension $3n$	Schoen
$2n \geq 8$, other	open, apart from special members	–

In every row marked open, the Weil classes are algebraic on a dense set of special members (Proposition 6.2), and the question is whether that set has interior. That is the subject of the last chapter.

Chapter 7

The remaining case

This chapter attempts the general case. Fix a Weil family $\pi: \mathcal{A} \rightarrow \text{Sh}$ of dimension $2n \geq 4$, as in Chapter 5, and a nonzero Weil class ω , extended to a flat global section by Proposition 5.6. By Corollary 5.9, the Hodge conjecture for a member with full Hodge group is the algebraicity of ω there. The attempt follows the most direct route: take a member where ω is algebraic and move the cycle to every other member. We prove what that route gives and show exactly where it stops. Then we look at the problem as a case of the variational Hodge conjecture and prove three results that narrow where a proof could come from.

Much of the material in the first sections first appeared in my preprint [BhH8]. Everything used is proved here.

7.1 The algebraic locus

Definition 7.1. The *algebraic locus* is the set $\Sigma \subseteq \text{Sh}$ of points t at which $\omega(t)$ is the class of an algebraic cycle with rational coefficients on A_t .

The goal is $\Sigma = \text{Sh}$.

Lemma 7.2. Σ is a countable union of closed algebraic subsets of Sh .

Proof. Fix a relatively ample line bundle on $\mathcal{A}/\text{Sh}$. For each Hilbert polynomial P the relative Hilbert scheme $\text{Hilb}_P(\mathcal{A}/\text{Sh})$ is projective over Sh and has finitely many irreducible components. Take finitely many components $H_1, \dots, H_r$, possibly from different Hilbert schemes, and rational numbers $q_1, \dots, q_r$. Over the fibre product $H = H_1 \times_{\text{Sh}} \dots \times_{\text{Sh}} H_r$ we have the class $\sum_i q_i [Z_i]$ in the fibre $H^{2n}(A_t, \mathbb{Q})$. Classes of the fibres of a flat family of subschemes are locally constant. So on each connected component of H the class $\sum q_i [Z_i]$ either equals $\omega(t)$ everywhere or nowhere. The union of the components where it equals $\omega(t)$ maps properly to Sh , so its image is closed and algebraic. There are countably many choices of polynomials, components and coefficients, and every point of Σ lies in one of these images. $\square$

Lemma 7.3 (Dichotomy). *Either $\Sigma = \text{Sh}$, or Σ is meagre: it lies in a countable union of closed subsets with empty interior, and its complement is dense.*

Proof. Sh is irreducible. If one of the closed sets of Lemma 7.2 is all of Sh , then $\Sigma = \text{Sh}$. Otherwise each of them is a proper closed algebraic subset, which has empty interior, and Baire's theorem on the complex manifold Sh gives the rest. $\square$

By Section 6.1, every Weil family has a member s_0 at which ω is a polynomial in divisor classes, and it can be chosen with complex multiplication [BhH8]. By Proposition 6.2 the isogeny orbit of s_0 lies in Σ and is dense. So Σ is dense, and the problem is entirely about which side of the dichotomy we are on. One stratum of Lemma 7.2 equal to all of Sh would settle it.

There is a catch, and it is the reason the problem is hard. Recall that Sh° is the set of points where the Hodge group is all of $\text{SU}(V, H)$. Its complement is meagre (Proposition 5.7). Every point isogenous to a CM point is a CM point, with a torus for its Hodge group, so the whole orbit of s_0 lies *outside* Sh° . To close the locus we need at least one point of $\Sigma \cap \text{Sh}^\circ$. For $K = \mathbb{Q}(\sqrt{-3})$ and a split form, Schoen's Prym varieties supply such points (Section 6.3). For other fields and discriminants no such point is known when $n \geq 3$.

7.2 The first route: strata with big monodromy

Suppose that we had found one point $t \in \Sigma \cap \text{Sh}^\circ$, and let S be an irreducible stratum of Lemma 7.2 through it. The first thing to ask is what S can look like. The following lemma says that S cannot be a small, degenerate piece of the family.

Theorem 7.4. *Let $S \subseteq \text{Sh}$ be an irreducible closed subvariety of positive dimension that contains a point of Sh° . Then the connected algebraic monodromy group of the local system $R^1\pi_*\mathbb{Q}$ restricted to the smooth locus of S is the whole group $\text{SU}(V, H)$.*

Proof. Let S^{reg} be the smooth locus of S . It is connected, since S is irreducible. The Hodge group of a very general point of S contains the Hodge group of every point of S , so it contains $\text{SU}(V, H)$. On the other hand every Hodge group in the family lies in $\text{SU}(V, H)$, by (5.2). So the generic Hodge group on S is $\text{SU}(V, H)$.

By André's theorem [And92], the connected algebraic monodromy group M of a polarisable variation of Hodge structure over a smooth connected base is a normal subgroup of the derived group of the generic Mumford–Tate group. Here that derived group is $\text{SU}(V, H)$, which is an outer form of SL_{2n} over $\mathbb{Q}$, and so it is almost simple. Its connected normal subgroups are 1 and itself.

If $M = 1$, the monodromy is finite. After a finite étale cover of S^{reg} the local system becomes trivial, and the period map lifts to a holomorphic map from that cover to D . Now D is a bounded domain, so the coordinate functions of the lift are bounded holomorphic functions. Pass to a resolution and a smooth compactification of the cover. By the Riemann extension theorem the bounded functions extend to the compactification, which is compact and connected, so they are constant. Then the period map is constant on S^{reg} . But the period map of the universal family over Sh is the identity of $\Gamma \backslash D$, so S is a point, which contradicts $\dim S > 0$. Hence $M = \text{SU}(V, H)$. $\square$

So a positive-dimensional stratum through a general point carries all the monodromy the family has. It is natural to hope that a cycle family with this much monodromy can be moved off S . That hope is exactly the variational Hodge conjecture for this family, and here is why the monodromy does not help on its own. The class ω is invariant under all of $\text{SU}(V, H)$, so the monodromy never moves ω , and it gives no constraint that would force the stratum to grow. Big monodromy rules out degenerate strata. It does not produce new cycles.

7.3 The second route: deforming one cycle

The direct way to move a cycle is deformation theory. Let $Z \subset A_t$ be a local complete intersection of codimension n . Bloch [Blo72] defined the *semiregularity map*

$$\sigma_Z: H^1(Z, N_Z) \longrightarrow H^{n+1}(A_t, \Omega^{n-1}),$$

and proved the following. If σ_Z is injective, then Z deforms along every deformation of A_t along which its class stays of Hodge type. Buchweitz and Flenner [BF03] extended the definition and the theorem to perfect complexes, and Pridham [Pri24] gave a general proof.

Theorem 7.5. *Suppose that at some point $t \in \text{Sh}$ there is a local complete intersection $Z \subset A_t$ of codimension n whose class is $\text{cl}(Z) = \alpha + c\eta^n$ with α a nonzero Weil class and $c \in \mathbb{Q}$, and whose semiregularity map is injective. Then $\Sigma = \text{Sh}$.*

Proof. Along the universal family over a small neighbourhood U of t , the flat transport of $\text{cl}(Z)$ is $\alpha + c\eta^n$, and both terms are Hodge classes at every point, because the Weil plane and η are. So the class of Z stays of Hodge type along every direction of U . By Bloch's theorem, Z deforms to every order along every curve germ in U . Then the image of its Hilbert scheme component contains the formal neighbourhood of t . Since that image is a closed analytic subset of U , it contains a neighbourhood of t . So one stratum of Lemma 7.2 has nonempty interior, and it is all of Sh by irreducibility. At every point s we then have an algebraic cycle of class $\alpha(s) + c\eta^n$, so $\alpha(s)$ is algebraic.

It remains to pass from α to ω . The field K acts on the Weil plane: an element $\tau \in K^\times$ acts on $\bigwedge_K^{2n} V$ by multiplication by τ^{2n} . A positive integer multiple $m\tau$ is a genuine endomorphism of A_s . Its pullback, divided by m^{2n} , is τ^* on H^{2n} , so τ^* carries algebraic classes to algebraic classes. Choose τ whose argument is not a rational multiple of π , for example $\tau = a + \sqrt{-d}$ with a a positive integer chosen so that $\tau/\bar{\tau}$ is not a root of unity. Then $\tau^{2n} \notin \mathbb{Q}$, and the classes α and $\tau^*\alpha$ span the Weil plane over $\mathbb{Q}$, so $\omega(s)$ is algebraic. $\square$

The same holds with Z replaced by a perfect complex E on A_t , with the Chern character in place of the cycle class, by the theorem of Buchweitz–Flenner and Pridham. This is the route Markman followed for sixfolds of split Weil type, using secant sheaves and an equivariant form of semiregularity [Mar25b].

7.4 Where the attempt stops

Both routes need an input that we do not have.

The first route needs a point of Σ with full Hodge group.

Outside the families where the conjecture is already known and Schoen's Prym locus for $\mathbb{Q}(\sqrt{-3})$, every point at which ω is known to be algebraic is a CM point or lies on a proper special subvariety. These points are excluded from Sh° by definition. Theorem 7.4 tells us what a stratum through a point of Sh° looks like, but it gives no way to find one. Where one is known, Schoen's Prym locus, it has dimension $3n$, while Sh has dimension n^2 . Nothing in the first route makes such a stratum grow.

The second route needs a semiregular object, and the obvious ones are not.

The first thing to try is an object whose Chern character is exactly a Weil class, with no other component. Voisin's tori rule this out.

Theorem 7.6. *Let $n \geq 2$, let $t \in \text{Sh}$, and let F be a perfect complex on A_t with $\text{ch}(F) = N\omega(t)$ for some $N \neq 0$, possibly plus components in degrees 0 and $4n$. Then the semiregularity map of F is not injective.*

Sketch of proof. Suppose it is injective. The class $N\omega$ is of Hodge type not only on the members of the Weil family but on every Weil torus of Section 5.5, which form a connected complex manifold $\mathcal{S}$ of dimension $2n^2$ containing Sh locally. So the theorem of Buchweitz–Flenner and Pridham [BF03, Pri24] kills every obstruction to deforming F along an Artinian deformation of A_t inside $\mathcal{S}$. Kuranishi's method then gives an analytic family of complexes of vector bundles over a neighbourhood of t in $\mathcal{S}$, deforming F . At a very general Weil torus T in that neighbourhood, every coherent sheaf has $\text{ch}_k = 0$ for $0 < k < 2n$, by the argument after Theorem 5.12. So ch_n of the deformed complex vanishes on T , while it is the flat transport of $N\omega \neq 0$. This is a contradiction. The details are written out in [BhH8]. $\square$

So a semiregular object has to carry a polarisation term, as in Theorem 7.5, and that term must not stay of Hodge type on the Weil tori, or the same argument applies. With a polarisation term the criterion survives, but it asks for an object with a precise and large amount of self-extension. For $n \geq 3$ and a general imaginary quadratic field no such object is known, and the natural constructions that provably fail are listed in [BhH8].

Neither gap is a technicality.

If $\Sigma = \text{Sh}$, then Σ meets Sh° , so the first gap is a consequence of the goal as well as a step towards it. And closing the locus is equivalent to one case of the Lefschetz standard conjecture: the case of the total space of the family over a projective curve through s_0 and a very general point [BhH8]. These are the variational Hodge conjecture and the standard conjectures in disguise. They are not small residuals.

7.5 The variational form of the problem

The question whether $\Sigma = \text{Sh}$ is one instance of the variational Hodge conjecture, and it helps to look at it that way. Let $f: X \rightarrow S$ be a smooth projective morphism onto a smooth connected variety, and let α be a global section of $R^{2p}f_*\mathbb{Q}$. By Deligne's theorem of the fixed part [Del71], the global sections carry a Hodge structure that does not depend on the point, and the inclusion into each fibre is a morphism of Hodge structures. So if $\alpha(s)$ is a Hodge class at one point, it is a Hodge class at every point. The variational Hodge conjecture asks for the corresponding statement about cycles: if $\alpha(s_0)$ is algebraic at one point s_0 , then $\alpha(s)$ is algebraic at every point s .

In our family, ω is a global section, it is a Hodge class everywhere, and by Proposition 6.2 it is algebraic at s_0 . The variational conjecture for $\pi: \mathcal{A} \rightarrow \text{Sh}$ and this one class says exactly that $\Sigma = \text{Sh}$. Three things are known about the conjecture in general, and none of them settles this case.

- (a) It holds when $p = 1$, by the theorem of Lefschetz on $(1, 1)$ classes. Here $p = n \geq 2$.
- (b) It follows from the Hodge conjecture for a smooth projective compactification $\bar{X}$ of the total space. By the global invariant cycle theorem [Del71] there is a class on $\bar{X}$ restricting to $\alpha(s)$, by semisimplicity of polarisable Hodge structures it can be chosen to be a Hodge class, and the restriction of a cycle representing it represents $\alpha(s)$ at every s . For our family $\bar{X}$ is a variety of dimension $2n + n^2$, or $2n + 1$ after restricting to a curve, and its Hodge conjecture is no easier than the one we want.
- (c) André's deformation principle [And96] says that a flat section which is motivated at one point is motivated at every point, and André also showed that every Hodge class on an abelian variety is motivated. So $\omega(t)$ is already known to be motivated for every t . A motivated class is algebraic as soon as the Lefschetz standard conjecture holds for the varieties that enter its construction. This is the same reduction to the standard conjectures as in Section 7.4, reached from the other side.

The rest of this section contains three results. The first changes the question into a question about degrees. It is a variant, with Chow varieties and arbitrary Zariski-dense sets, of the bounded criterion of [BhH8], which is stated for Hilbert data along isogeny orbits. The second shows that nothing built from divisors reaches ω at a member with full Hodge group. Its proof is the argument that [BhH8] gives for Mumford's exceptional classes, carried over to the Weil family. The third is the analogue for the tautological cycles of curves with an automorphism of order three.

7.5.1 A criterion in terms of degrees

Measure the size of a cycle by its degree against the polarisation: for an effective cycle Z of dimension n on A_t , put $\deg_\eta Z = \int_{A_t} \text{cl}(Z) \cup \eta^n$. Every cycle with rational coefficients can be written as $\frac{1}{m}(Z^+ - Z^-)$ with $Z^\pm$ effective and m a positive integer. For $N \geq 1$ let Σ_N be the set of points t at which

$$\omega(t) = \frac{1}{m}(\text{cl}(Z^+) - \text{cl}(Z^-))$$

for some effective cycles $Z^\pm$ on A_t of dimension n with $\deg_\eta Z^\pm \leq N$ and some integer $1 \leq m \leq N$.

Proposition 7.7. *Each Σ_N is a closed algebraic subset of Sh , and $\Sigma = \bigcup_N \Sigma_N$. The following are equivalent.*

- (i) $\Sigma = \text{Sh}$.
- (ii) $\Sigma_N = \text{Sh}$ for some N .
- (iii) For some N , the set Σ_N contains a Zariski-dense subset of Sh .

Proof. Effective cycles of dimension n and bounded degree in the fibres of the projective family $\mathcal{A} \rightarrow \text{Sh}$ are parametrised by a relative Chow variety which is of finite type and proper over Sh [Kol96, Ch. I]. Over a connected component of it, the class of the cycle is a flat section of $R^{2n}\pi_*\mathbb{Q}$, so the condition that $\frac{1}{m}(\text{cl} Z^+ - \text{cl} Z^-) = \omega$ holds on whole connected components of the fibre product of two such Chow varieties, or nowhere on them. Taking the union over

$m \leq N$ of the components where it holds and projecting to Sh gives Σ_N , which is therefore closed and algebraic. Every point of Σ lies in some Σ_N .

(i) implies (ii): Sh is the countable union of the closed sets Σ_N , so by Baire's theorem one of them has nonempty interior, and since Sh is irreducible that one is all of Sh . (ii) implies (iii) trivially. (iii) implies (i): a Zariski-closed set that contains a Zariski-dense set is everything. $\square$

The proposition is elementary, but it changes the shape of the question, as [BhH8] already observed in its own setting. Proposition 6.2 already gives algebraic representatives of ω on a dense set. What is missing is a *uniform bound* on their degrees and denominators along some Zariski-dense subset. Transport along isogenies does not give one, and it is worth seeing why.

Remark 7.8. Let $\phi: A_t \rightarrow A_{t'}$ be an isogeny of degree d that commutes with K and satisfies $\phi^*\eta' = e\eta$ for an integer $e \geq 1$, as the Hecke correspondences do. The two polarisations have the same type, so comparing Euler characteristics gives $e^{2n}\chi(\eta) = \chi(\phi^*\eta') = d\chi(\eta') = d\chi(\eta)$, that is $d = e^{2n}$. For an effective cycle Z of dimension n the projection formula gives

$$\deg_{\eta'} \phi_* Z = \int_{A_t} \text{cl}(Z) \cup \phi^*\eta'^n = e^n \deg_{\eta} Z.$$

An abelian variety has only finitely many isogenies of bounded degree, so along any infinite subset of the Hecke orbit of s_0 the multipliers e are unbounded, and so are the degrees of the transported cycles. Proposition 7.7 asks for *other* cycles at those points, of bounded degree. The Hecke construction gives no hint where to find them.

7.5.2 A barrier at members with full Hodge group

Fix $t \in \text{Sh}^\circ$, write $A = A_t$, and let $U = U(V, H)$ be the unitary group of the hermitian form, as an algebraic group over $\mathbb{Q}$. It acts on $V = H^1(A, \mathbb{Q})$, and so on $H^*(A, \mathbb{Q}) = \bigwedge^* V$. Let $\mathcal{C}$ be the category of abelian varieties isogenous to a power of A , with homomorphisms as morphisms. Choosing an isogeny $B \rightarrow A^m$ lets U act on $H^1(B, \mathbb{Q}) \cong V^m$ diagonally. Two choices differ by an invertible element of $\text{End}^0(A^m)$, and we shall see in a moment that these commute with U , so the action does not depend on the choice.

Theorem 7.9. *Let $n \geq 2$ and $t \in \text{Sh}^\circ$. Let $(\mathcal{L}(B))_{B \in \mathcal{C}}$ be the smallest family of subspaces $\mathcal{L}(B) \subseteq H^*(B, \mathbb{Q})$ such that each $\mathcal{L}(B)$ is a subalgebra containing the classes of all divisors on B , and $h^*\mathcal{L}(B') \subseteq \mathcal{L}(B)$ and $h_*\mathcal{L}(B) \subseteq \mathcal{L}(B')$ for every homomorphism $h: B \rightarrow B'$ in $\mathcal{C}$. Then every class in every $\mathcal{L}(B)$ is invariant under U . In particular no class in $\mathcal{L}(A)$ has the form $\alpha + c\eta^n$ with α a nonzero Weil class.*

Proof. Step 1: the endomorphisms. Over $\mathbb{C}$ the space V splits as $V_+ \oplus V_-$, the two eigenspaces of K , each of dimension $2n$. The polarisation form E is invariant under the norm-one elements of K , which forces V_+ and V_- to be isotropic and E to pair them perfectly. So, as representations of $\text{SU}(V, H)_{\mathbb{C}} \cong \text{SL}_{2n}$, they are the standard representation and its dual. For $2n \geq 3$ these are irreducible and not isomorphic, so the commutant of $\text{SU}(V, H)$ in $\text{End}_{\mathbb{Q}}(V)$ has dimension two. Since the Hodge group of A is $\text{SU}(V, H)$, this commutant is $\text{End}^0(A)$, and so $\text{End}^0(A) = K$ and $\text{End}^0(A^m) = M_m(K)$. A homomorphism $h: A^m \rightarrow A^k$ therefore acts on H^1 by a matrix with entries in K , and since U acts K -linearly on V , the map $h^*: V^k \rightarrow V^m$ commutes with the diagonal action of U . The same holds in $\mathcal{C}$ after composing with isogenies.

Step 2: the centre. The centre of U is the torus $U(1)_K$ of norm-one elements of K , and $U = \mathrm{SU}(V, H) \cdot U(1)_K$. An element λ of the centre acts on V_+ by λ , through the fixed embedding of K , and on V_- by λ^{-1} , so on the summand $\bigwedge^a V_+^{\oplus m} \otimes \bigwedge^b V_-^{\oplus m}$ of $H^{a+b}(A^m, \mathbb{C})$ it acts by λ^{a-b} . Hence a class is U -invariant exactly when it is $\mathrm{SU}(V, H)$ -invariant and lies in the sum of the summands with $a = b$.

Step 3: divisors. A divisor class on A^m is a Hodge class, and the Hodge group of A^m is $\mathrm{SU}(V, H)$ acting diagonally, so the class is $\mathrm{SU}(V, H)$ -invariant. Its components with $a \neq b$ lie in $\bigwedge^2(V_+ \otimes \mathbb{C}^m)$ or its conjugate. As a representation of SL_{2n} this is a sum of copies of $\bigwedge^2 V_+$ and $\mathrm{Sym}^2 V_+$, which have no invariants when $2n \geq 3$. So those components vanish, and by Step 2 the class is U -invariant.

Step 4: the operations. Cup product and h^* commute with U , by Step 1. Every element of U preserves E , so it has determinant one on V and acts trivially on the top cohomology of each B . So U preserves the Poincaré pairing, and h_* , being the adjoint of h^* for that pairing, commutes with U as well. It follows that the family of U -invariant subalgebras $H^*(B, \mathbb{Q})^U$ has all the properties in the statement, and by minimality $\mathcal{L}(B) \subseteq H^*(B, \mathbb{Q})^U$.

Step 5: Weil classes. The Weil plane, tensored with $\mathbb{C}$, is $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$, the summands with $(a, b) = (2n, 0)$ and $(0, 2n)$. The centre acts on them by $\lambda^{\pm 2n}$. Taking $\lambda = \tau/\bar{\tau}$ with τ as in the proof of Theorem 7.5, this is not 1, so the Weil plane has no nonzero U -invariant vector. Since η^n is U -invariant, a U -invariant class $\alpha + c\eta^n$ has α U -invariant, hence $\alpha = 0$. $\square$

The theorem reaches further than it looks, because Chern characters behave well under the same operations.

Corollary 7.10. *Let $t \in \mathrm{Sh}^\circ$, and let F be an object of the derived category of A_t obtained from line bundles on varieties in $\mathcal{C}$ by finitely many shifts, direct sums, cones, derived tensor products, and derived pullbacks and pushforwards along homomorphisms. Then $\mathrm{ch}(F)$ is U -invariant, and F cannot satisfy the hypothesis of Theorem 7.5. This applies in particular to Fourier–Mukai transforms whose kernel is the Mumford line bundle $m^*L \otimes \mathrm{pr}_1^*L^{-1} \otimes \mathrm{pr}_2^*L^{-1}$ on $A_t \times A_t$, and to pushforwards of line bundles along isogenies, which by a theorem of Mukai include all simple semi-homogeneous vector bundles [Muk78].*

Proof. The Chern character is additive on exact triangles and multiplicative on derived tensor products, it commutes with pullback, and $\mathrm{ch}(L) = \exp c_1(L)$ for a line bundle. For pushforward along a homomorphism h , the Grothendieck–Riemann–Roch theorem gives $\mathrm{ch}(Rh_*F) = h_*\mathrm{ch}(F)$, because the Todd class of an abelian variety is 1. So $\mathrm{ch}(F)$ lies in $\mathcal{L}(A_t)$, and Theorem 7.9 applies. $\square$

So at a member with full Hodge group, nothing built from line bundles and homomorphisms has the right Chern character. This explains a feature of every successful construction so far. At s_0 the endomorphism algebra is large, the corresponding group is smaller than U , and ω is a polynomial in divisor classes. Markman’s secant sheaves also start on special members, built from an abelian n -fold and its dual, and reach the general member only by deformation [Mar25a, Mar25b]. The theorem shows that this detour is forced: a proof has to start where the endomorphisms are large and then deform, or it has to find at a general member a cycle that does not come from divisors at all.

7.5.3 Curves with an automorphism of order three

The oldest source of cycles that do not come from divisors is curves. For $K = \mathbb{Q}(\sqrt{-3})$ there are two natural families of curves whose Jacobians contain abelian varieties of Weil type, and one might hope that cycles built from the curve reach ω . The next proposition shows that the obvious cycles do not.

Let C be a smooth projective curve with an automorphism ρ of order three, and let $V \subseteq H^1(C, \mathbb{Q})$ be the kernel of $1 + \rho^* + \rho^{*2}$, the part on which ρ has no fixed vectors. Then K acts on V through ρ . Let $P \subseteq J(C)$ be the image of $1 - \rho$, so that $H^1(P, \mathbb{Q})$ is identified with V , and assume that P , with the polarisation induced from $J(C)$, is of Weil type for K . Let $U = U(V, H)$ act on $H^1(C, \mathbb{Q}) = V \oplus H^1(C, \mathbb{Q})^\rho$, trivially on the second summand, and trivially on H^0 and H^2 .

Proposition 7.11. *Let $R^*(C^m) \subseteq H^*(C^m, \mathbb{Q})$ be the subring generated by the classes of the diagonals $\{x_i = x_j\}$, the graphs $\{x_j = \rho^k(x_i)\}$ for $k = 1, 2$, and the pullbacks of the point class from each factor. Let $f: C^m \rightarrow P$ be a morphism of the form $h \circ (\text{AJ} \times \cdots \times \text{AJ})$, where AJ is an Abel–Jacobi map $C \rightarrow J(C)$ and $h(y_1, \dots, y_m) = (1 - \rho) \sum_i \psi_i(y_i)$ with $\psi_i \in \mathbb{Z}[\rho]$. Then every class in $f_* R^*(C^m)$ is U -invariant, and in particular none has the form $\alpha + c\eta^n$ with α a nonzero Weil class.*

Proof. The summand V is orthogonal to $H^1(C, \mathbb{Q})^\rho$ under cup product, because the pairing is ρ -invariant and ρ acts on the two summands with no common eigenvalue. The group U preserves the pairing on V , which is the polarisation form of P up to a constant. So the action of U on $H^1(C, \mathbb{Q})$ is symplectic, it extends to an action on $H^*(C^m, \mathbb{Q})$ by algebra automorphisms, and it fixes the top class.

The class of the graph of an endomorphism T of H^1 , in the $H^1 \otimes H^1$ part of $H^2(C \times C)$, is the tensor corresponding to T under Poincaré duality. It is invariant under every symplectic g that commutes with T . The diagonal is the case $T = 1$, and the graphs of ρ^k are the cases $T = \rho^{\pm k}$, which commute with U because U is K -linear. The other components of these classes, and the point classes, lie in $H^0 \otimes H^2$ and $H^2 \otimes H^0$, where U acts trivially. So $R^*(C^m)$ is U -invariant.

The pullback $f^*: H^1(P) = V \rightarrow H^1(C)^{\oplus m}$ is given in each coordinate by an element of $\mathbb{Z}[\rho]$, that is, by multiplication by an element of K . So it lands in $V^{\oplus m}$ and commutes with U , which acts K -linearly. (This is where the special form of h is used: a general K -linear map of V does not commute with U .) Hence f^* commutes with U on all of $H^*(P)$, and f_* , its adjoint for the Poincaré pairings, does too. The last assertion follows as in Step 5 of the proof of Theorem 7.9. $\square$

Two families fit the hypotheses. In the first, C is the cyclic triple cover $y^3 = \prod_{i=1}^{n+1} (x - a_i) \prod_{j=1}^{n+1} (x - b_j)^2$ of the projective line. By Riemann–Hurwitz it has genus $2n$, and by the Chevalley–Weil formula the two nontrivial eigenspaces of ρ on holomorphic differentials both have dimension n , so $P = J(C)$ is of Weil type. These curves depend on $2n - 1$ parameters, which is less than n^2 for $n \geq 2$, so their Jacobians fill only a proper subvariety of Sh . In the second, $C \rightarrow C'$ is an étale cyclic triple cover of a curve of genus $n + 1$. Then C has genus $3n + 1$, the two nontrivial eigenspaces both have dimension n , and P is the Prym variety, of dimension $2n$. These depend on $3n$ parameters, and for $n \leq 3$ that is at least n^2 . Here much more is known than the proposition might suggest. Schoen proved that the Weil classes of these Prym varieties are algebraic [Sch88], so for $n \leq 3$ the Pryms give whole families of Weil

type in which the conjecture holds, and for $n \geq 4$ they give a locus of dimension $3n$ inside Sh on which ω is algebraic and whose very general point has full Hodge group. The proposition says something about how this can happen. Schoen's cycles represent a nonzero Weil class, so they cannot be the image of a tautological class under a map of the form in the proposition. Whatever they are built from, it is a cycle on a power of C , or a map, outside the tautological world. That is exactly the kind of source that Theorem 7.9 says a proof needs, and it is the most concrete lead this chapter can offer (see Section 7.6).

7.6 What is left

Put together, the three results narrow the search. At a member with full Hodge group, ω cannot be reached by anything built from divisors and homomorphisms (Theorem 7.9 and Corollary 7.10), nor by the tautological cycles of a curve with an automorphism of order three (Proposition 7.11). At special members it can be reached, and Hecke transport spreads those representatives over a dense set, but with degrees that the transport itself drives to infinity, and Proposition 7.7 says that bounded degree on a Zariski-dense set is exactly what is needed. Any proof therefore has to supply one of three things: a uniform bound on degrees along a Zariski-dense set of special members; a deformation argument that carries a cycle from a special member to the general one, which is the semiregularity route of Theorem 7.5; or a source of cycles at a general member that is neither divisorial nor tautological. The third exists for $\mathbb{Q}(\sqrt{-3})$: Schoen's cycles on Prym varieties. Their locus has dimension $3n$, so for $n \geq 4$ they reach only a thin part of Sh . The obvious question is whether one of Schoen's cycles is semiregular. The construction makes the question concrete. On the symmetric power $C'^{(2n)}$ of the base curve, the canonical system $|K_{C'}| \cong \mathbb{P}^n$ is the fibre of the Abel–Jacobi map over the canonical class. The triple cover defines an étale cover of $C'^{(2n)}$ of degree three, and over $|K_{C'}|$, which is simply connected, it splits into three copies of $\mathbb{P}^n$. The differences of the copies, weighted by the characters of order three, are Schoen's cycles [Sch88, Theorem 2.0 with $r = 0$], and Patel and Zhang reach the same copies by base change along an isogeny onto the Jacobian [PZ25, Theorem 1.1]. On the Prym the class of one copy is a Weil class plus a multiple of η^n . When that multiple is nonzero, this is exactly the shape that Theorem 7.5 allows and that the obstruction from Weil tori does not exclude. If one of them were semiregular, Theorem 7.5 would give $\Sigma = \text{Sh}$ for the family of that dimension. The construction also suggests why this may fail. The Abel–Jacobi curve $C \subset J(C)$ comes from the same map, and its class $\theta^{g-1}/(g-1)!$ is algebraic on every principally polarized abelian variety; yet by the criterion of Matsusaka and Ran a curve with this class exists only on Jacobians and products of Jacobians [Mat59, Ran81]. For $g \geq 4$ these fill a proper subvariety of the moduli space, so by Bloch's theorem the Abel–Jacobi curve is not semiregular. Schoen's cycles are the analogue one step up, and nothing in their construction moves them off the Prym locus. This is an analogy, not a proof, and I have not been able to decide the question. Outside $\mathbb{Q}(\sqrt{-3})$, none of the three is available for $n \geq 3$ in general.

To sum up what this chapter proves. The algebraic locus of the Weil class is a countable union of closed algebraic subsets, it is dense, and it is either everything or meagre. A positive-dimensional stratum of it through a member with full Hodge group has full monodromy. A single semiregular cycle or complex, at any member, with class a nonzero Weil class plus a multiple of η^n , would close the locus, while an object whose Chern character is a pure Weil class is never semiregular. The problem is the variational Hodge conjecture for one class. It is

equivalent to a bound on degrees along a Zariski-dense set, and at a member with full Hodge group it cannot be solved with divisors or with the tautological cycles of curves.

What it does not prove is the conclusion. For $n \geq 4$, and for $n = 3$ outside the known families, the Hodge conjecture for abelian varieties of Weil type remains open. Even a complete solution would leave the other exceptional classes on abelian varieties, and the conjecture for varieties that are not abelian, which is the conjecture in full.

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